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The risk of infection and contribution to transmission of SARS-CoV-2 in school staff - a systematic review

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ABSTRACT

Introduction: The comparative risk of infection in school staff and the comparative contribution of school staff to SARS-CoV-2 transmission dynamics remain unclear.

Methods: We conducted a rapid systematic review of studies that reported the risk of SARS-CoV-2 infection in school staff or the transmission dynamics of SARS-CoV-2 in school settings. We searched MEDLINE, WHO COVID database and preView with a sensitive compound search strategy. Data synthesis was qualitative and we report attack rates and infection risk in school settings for staff and students stratified by control measures taken and infection dynamics at the point of data collection.

Results:

18 studies from 1794 identified records were included.

We found low attack rates in three studies in a low infection incidence setting, similar for teachers and students. Five studies conducted during medium incidence settings showed up to 4.4% secondary infections following index cases. Two studies in a high population infection settings showed that up to 16% of school staff had secondary infections.

Regarding seroprevalence studies, two studies show an infection risk of 0 and 0.2% for teachers with a medium incidence of infection. Following a high incidence of infection, two studies demonstrated a seroprevalence of 1.7 and 28%.

In studies comparing risk of infection in school staff to other groups, the risk of infection for teachers in a high infection environment is higher in face-to-face than in distance classes when compared with general population groups. The risk of infections as well as risk of hospitalization both increased for teachers during school openings compared to school closure.

Conclusion: While in low incidence settings there is little evidence for school staff to be at high risk of SARS-CoV-2 infection, in high incidence settings there is an increased risk of SARS-CoV-2 infection in school staff teaching face-to-face compared to staff teaching digitally and general population.

Key words: SARS-CoV-2, attack rate, infection risk, school staff, students

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Article Summary

Strength and Limitations of this study:

- The study results were stratified according to the prevalence of infection during data collection period and prevailing control measures in the school setting at that time.
- The infection risk in teachers/school staff were compared with infection risk in students, general population, and teachers.
- The results from included studies were heterogeneous.

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INTRODUCTION

Corona virus disease-19 (COVID-19) is a global public health threat, caused by severe acute respiratory syndrome coronavirus-2 (SARS-CoV-2).[1] Although people of all ages are affected, the severity of the clinical course increases with age (more severe in people >65 years of age).[2, 3] Children and adolescents most commonly experience a mild clinical course and show less severe outcomes compared to adults and ageing people.[4-7] When showing severe outcomes, long-term complications can be equal or worse in children than in adults.[8]

Non-pharmaceutical interventions (NPIs) like isolation, quarantine and social distancing including large-scale school closures are applied near-universally to curb the transmission of SARS-CoV-2.[9, 10] Such conventional public health measures appear to reduce the number of new infections.[10, 11] However, school closures alone are not sufficient to prevent community transmission of SARS-CoV-2.[12, 13]

Several systematic reviews, meta-analyses and large ecological analyses have focused on effects and adverse effects of school closures mainly assessing endpoints concerning the effect on community transmission as well as effects on children.[9, 14, 15] Long-term school closures are a threat to the physical and mental health of children and adolescents and intensify the racial and socioeconomic gaps in society.[16-20]

Nevertheless, keeping schools open when community transmissions are increasing may be posing a threat to school staff in particular, as their age leaves them more at risk of severe infections compared to students. Evaluating the risk to school staff as well their role in schools and community transmission is thus essential to an evidence-based approach to pandemic public health strategies.

In an umbrella review (Lange et al, submitted) we did not find any systematic review focusing on risk of and contribution to transmission of school staff.

The risk of infection in school staff in dynamic infection environments depends on the population infection dynamic as well as the infection dynamic within schools, the susceptibility of staff to the infection and the number of contacts of the staff at that time. An absolute estimate of the risk of infection is futile due to its dependence on the evolving context. We have therefore collated the existing evidence on the relative risk of infection compared to other population groups in original papers and existing reports and stratified by infection dynamic prevalent during the period of data collection.

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METHODS

Protocol and registration

We followed PRISMA guidelines[21] and registered this review with PROSPERO (CRD42021239225).

Search strategy

We searched MEDLINE and preView most recently on 29th January 2021 by using search terms "SARS-CoV-2"/ "COVID-19" and "teacher"/"school" combinations with OR and AND Boolean operators. We also searched WHO COVID-19 database for relevant literatures.

We did not restrict our search to any study design or language of publication. Preprints are included in this search if available from preprint databases. We did not consider the preprint available only on homepages or institute websites.

Eligibility criteria

Studies reporting the risk of COVID-19 in teachers or any school staff or any kind of involvement of teacher or school staff in SARS-CoV-2 transmission were eligible for inclusion in the review. Articles published in peer-reviewed journals, preprints, technical reports, and case reports were included. Studies and reports were also included based on expert suggestion.

Modelling studies, opinion analysis, media reports, reviews, and meta-analysis were excluded. We also excluded studies reporting SARS-CoV-2 transmission in students and school staff but in different school settings and studies reporting solely risk factors for SARS-CoV-2 infection in teachers or students. The PICOs for the included studies is presented in table 1.

Table 1 PICOs for Included studies

Transmission of SARS-CoV-2 in school	Population	Intervention/ Exposure	Comparator	Outcome
	School staff, Any contacts of school staff	School, primary school, secondary school	School children, general population, present in the school (distance learning), school staff in different school forms or learning situations	Secondary attack rates, relative/infection risk, odds ratio,

Study selection

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Two reviewers (SJK and AJ) screened the title and abstracts and read the full-text independently based on the predefined eligibility criteria. Inconsistencies and disagreements in the judgement were resolved by consultation with a third reviewer (BL).

Patient and Public Involvement

Patients or the public were not involved in any stage of this systematic review.

Data Extraction

Two reviewers (SJK and AJ) independently extracted the data from included studies into a pre-specified form. Disagreements in the data extraction process were resolved by consultation with the third reviewer (BL). Data related to study characteristics (source, name of first author, study design/ type, date of data collection), study population (population of staff, population of students and population of contacts), main issue, study setting, comparator, attack rate in staff, attack rate in student, infection risk in student, infection risk in staff, outcome and results were extracted.

Quality appraisal

The Agency for Healthcare Research and Quality (AHRQ) checklist was adapted to assess the quality of included studies. [22-24]

The risk of bias domains used were selection bias, performance bias, attrition bias, detection bias, reporting bias and information bias. The overall risk of bias for included studies was classified as high, unclear, medium or low risk of bias. The criteria for *high risk of bias* for included studies are high risk of bias in any one of the domains. Studies with unclear risk of bias in any one of the key domains and no high risk of bias in any other domain were deemed *unclear risk of bias.* Studies with *medium risk of bias* in any one of the domains and low risk of bias in all other domains were deemed *medium risk of bias* and those with *low risk of bias* in all the domains were deemed *low risk of bias.*

Synthesis of results

Qualitative data synthesis was performed by describing study characteristics and main research questions, with the main conclusions of included studies presented narratively and in table format. The findings were presented based on the different type of SARS-CoV-2 transmission found in the school setting. When absolute numbers were available, we calculated (secondary) attack rates. When authors already calculated the attack rate we report them as given. Where infections risk is given by either seroprevalence or PCR based testing, we report them as given. All outcomes are reported stratified by infection environment and NPIs measures in place during data collection periods.

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RESULTS

Study selection and study characteristics:

The search yielded 1784 studies. Of these, eight met the inclusion criteria. A further 10 studies were found through screening references of systematic reviews, meta-analyses and following expert suggestions. Eighteen studies were included in the review; the selection process is described in Figure 1.

Almost all of the included studies were conducted in 2020. Ten of the included studies had a data collection/analysis period from January to June.[25-34] Three studies have collected data from April to July,[35] in June/July,[36] or in July only.[37] One study collected data from July to September [38] and two studies had data collection periods from August to November.[39, 40] One study uses data from March 2020 to January 2021,[41] and the remaining study analysed data from March 12, 2020.[42] During the data collection period, the total number of SARS-CoV-2 cases in the countries of study ranged from 1.44 cases/million to 26,802 cases/million and SARS-CoV-2-related deaths from 0.03 deaths/million to 339.68 deaths/million.[43] Similarly, the number of new cases per day at the start of the study interval ranges from 0 to 169.71 cases per million per day. At the end of the study period the incidence ranges from 0.29 to 423.22 cases per million per day.[43]

The studies were originated in Australia,[30, 35, 38] France,[27, 28] Germany,[25] Ireland,[42] Israel,[32] Italy,[40] Panama,[29] Singapore,[34] Sweden,[31, 33] the UK,[36] Scotland, [41] and the USA.[26, 37, 39] Among them three were reports published by the Public Health Agency of Sweden[31] and by the National Centre for Immunization Research and Surveillance (NCIRS), New South Wales, Australia.[35, 38] Five were published in preprint [25, 28, 33, 37, 41] with the remaining ten studies published in peer-review journals.

All included studies provide information about either risk of SARS-CoV-2 infection in teachers and/or students, transmission of SARS-CoV-2 in school settings or seroprevalence of SARS-CoV-2 IgG antibodies in school staff or school settings. Most of the included studies report attack rates,[27, 29, 30, 32, 34-36, 38-40, 42] seroprevalence of SARS-CoV-2 IgG [25, 26, 28, 37] or infection risk [31, 33, 41] among teachers and students. The characteristics of included studies are tabularized in table 2.

Quality Assessment:

Among the included studies, six have a low risk of bias, ten studies have a medium risk of bias, one has an unclear risk of and one has a high risk of bias. Reasons for assigning medium risk of bias were: not all contacts were tested in contact tracing studies and some of the studies used only case notification data from nationwide surveillance database. This increases the possibility of missing asymptomatic cases and cases that were not reported. The reason for assigning unclear risk of bias was that the occupation code was missing for 25% of the confirmed cases. It was unclear how the study group dealt

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with this issue. The reason for assigning high risk of bias to one of the studies was selective reporting of results, low participation rate and use of questionnaires to assess symptoms. This increases the possibility of recall and misclassification bias. The overall risk of bias assessments for included studies are tabularized (supplementary table 1).

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Table 2 Characteristics of Included studies

Source	First author	Region	Main issue	Type of study	Population	Population Staff	Population Contacts	Setting	Comparator	Date collection period	Outcome
Attack rate/secondary infections in school											
Lancet Child and Adolescent Health	Kristine Macartney [30]	Australia	Transmission of SARS-CoV-2 among children and staff in schools and early childhood education and care (ECEC) settings	Perspective cohort study	Total= 1475 27 primary infections (12 children and 15 adults) and 1448 contacts.	NA	1448	15 schools and 10 ECEC in New South Wales Australia 633 had nucleic acid or antibody testing or both	Students	January 25-1 May 2020	Attack rate
Euro Surveillance	Chen Stein-Zamir [32]	Israel	COVID-19 outbreak in school in Israel. The school hosts grades 7-12.	Observational	Total= 1314 2 primary infections among students. 1161 student contacts and 151 staff in the school	151	1312	SARS-CoV-2 outbreak in school in Israel after school reopening.	Students	May-June 2020	Attack rate
Euro Surveillance	Elisabetta Larosa [40]	Italy	Secondary transmission of COVID-19 cases in school and preschool setting in northern Italy	Observational	Total= 1248 48 index cases (43 students and 5 teachers) and 1200 contacts of primary cases.	209	1200	Total 36 schools and preschools in northern Italy	Students of different school levels	Sep 1- Oct 15	Attack rate
Frontiers of Public Health	Augusto Hernandez [29]	Panama	Secondary transmission of school staff	Case report	Total= 202 2 primary cases (1 teacher and 1 director) Approx. 200 contacts in school Household contacts of primary cases	NA	200	School in Panama	Students	February-March 2020	Secondary attack rate, attack rate
Euro Surveillance	Heavey [42]	Ireland	Secondary transmission in school setting	Observational	**Total in school = 1031** Total including other setting = 1161 (6 cases= 3 paediatric and 3 adult)	NA	**In school=1031** (1155 contacts including other settings)	School in Ireland	NA	12 March 2020	Secondary attack rate, attack rate
Clinical Infectious Diseases	Yung, CF [34]	Singapore	Secondary transmission in school setting	Observational	Total = 122 (3 primary cases (1 student from secondary school, 1 student from preschool and 1 adult staff from other preschool)	NA	119 (8+34+77)	3 different schools in Singapore	NA	Feb-March 2020	Secondary transmission, attack Rate
Clinical Infectious Diseases	Danis, K [27]	France	Tertiary transmission of SARS-CoV-2 in school setting	Observational	113 (1 secondary cases and 112 school contacts)	NA	112	3 different schools and 1 ski club in the French Alps	Teachers, students	Feb 2020	Tertiary transmission
National Centre for Immunisation Research and Surveillance, NSW, Australia	NCIRS [35]	Australia	Secondary transmission in different school settings	Observational	Total=527 6 primary cases (4 students and 2 staffs) and 521 close contacts from 6 different educational settings (5 schools and 1 ECEC)	62	521	6 different schools (5 schools and 1 ECEC) in NSW; Australia. The primary cases were found in the school setting and contact tracing was done and diagnostic test were performed in close contacts	Students	10 April -3 July	Secondary transmission, attack rate
National Centre for Immunisation Research and Surveillance, NSW, Australia	NCIRS [38]	Australia	Secondary transmission in different school settings	Observational	Total= 3857 39 primary cases (32 students and 7 staff) and 3824 close contacts in 34 different educational settings (28 schools and 6 ECEC)	385	3824	39 primary cases were found in 34 different educational settings (28 schools and 5 ECEC) in NSW Australia.	Students	4 July - 25 September	Secondary transmission, attack rate
Lancet Infectious Diseases	Ismail, SA [36]	UK	Estimating the rate of SARS-CoV-2 infection and outbreaks among staff and students in educational settings during the summer half-term (June–July, 2020) in England	Prospective cross sectional analysis	Students and staff in school of England	NA	NA	Schools in England	Students	June/July 2020	Secondary transmission
MMWR	Falk, A [39]	USA	Secondary transmission in the school setting	Observational	Total= 5530 184 primary cases with 5346 close contacts in 17 different educational settings in Wisconsin, USA. (4876 students and 654 staff in face-to-face learning)	654	5346	17 different schools settings; 8 elementary schools (k-6) and 9 secondary schools with grade 7 to 12 in Wisconsin, USA	Student	August 31 – November 29	Secondary transmission and attack rate
Infection risk in Teachers											
Public Health Agency Sweden	Public Health Agency Sweden [31]	Sweden	Risk of SARS-CoV-2 infection in teachers in Sweden	Observational Report	364760 (age 44-53) Teachers working in different school in Sweden	NA	NA	Different school settings in Sweden	Teachers of different school levels	March-May 2020	Relative risk

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medRvix	Jonas Vlachos [33]	Sweden	Risk of SARS-CoV-2 infection in parents, high school teachers, lower secondary school teacher and their partners were assessed after the partial school closure	Retrospective data analysis	Upper secondary and lower secondary school teachers, their partners and students' parents from Sweden	NA	NA	Upper secondary and lower secondary schools of Sweden	Teachers of different school levels and their partners. Parents of children attending different schools	June 15	Odds ratio
medRvix	Lynda, Fenton [41]	Scotland	Risk of COVID-19 and risk of hospitalization with COVID-19 among teachers compared with healthcare workers and general population of working age	Case control study	Total study population N= 871568 N teachers = 18479 N healthcare workers= 35461	18479	NA	Teachers and healthcare workers in Scotland	Health care workers, household members of healthcare workers and teachers	March 2020-January 2021	Risk rate
Seroprevalence studies											
Emerging Infectious Diseases	Nicole E Brown [26]	USA	Teacher to student transmission	Cross sectional (seroprevalence survey)	Total = 121 1 infected teacher and 120 student contacts	NA	120	School in the USA. All students were instructed to quarantine. 21 students participated in serological survey 14 days after quarantine	Students	Blood sample collection date: March 13, 2020	Secondary infection
medRvix	Armann, JK [25]	Germany	Seroprevalence study in teachers and students	Cross sectional (Seroprevalence study)	2045 participants (1538 students and 507 teachers)	507	NA	13 different secondary schools in Saxony, Germany	Students	Blood sample collection 25th May and 30th June	Seroprevalence
medRvix	Liah Lopez [37]	USA	Seroprevalence of anti-SARS-CoV-2 IgG antibodies in school staff in Midwestern USA	Population based seroprevalence study	1261 eligible school staff	753 staff members participated in the survey	NA	18 year or older staff of Lake Central School Corporation located in suburban Indiana employed in academic year 2018/19 or 2019/20 and with record pf annual wellness check	General population	July 2020	Seroprevalence of antibody
medRvix	Fontanet,A [28]	France	Assessing infection attack rate using serological assays	Observational cohort (Seroprevalence study)	661 (pupils, parents, siblings, including teachers and non-teaching staff)	661	NA	Secondary school and household in Oise department in Northern France	Students and general population	30 March- 4 April	Infection attack rate

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Findings

Attack rates in school staff and students

Eleven studies reported data on attack rates in schools.[27, 29, 30, 32, 34-36, 38-40, 42] The detail information is provided in supplementary table 2.

Four studies found no secondary infections in schools following index cases.[27, 34, 35, 42] Of the remaining seven studies, six reported attack rates of 0 to 13% following outbreaks among students and attack rates of 0 to 16.6% following outbreaks among school staff.[29, 30, 32, 38-40] One of the studies reports 100 secondary cases in staff and 22 in students related to one outbreak.[36]

Secondary attack rates among pupils were 0.14%,[39] 0.3%,[30] 0.81%,[38] and 3.8%.[40] The latter study further differentiated between 6.6% in secondary schools and 0.38% in primary schools with no secondary transmission in pre-schools.[40] The secondary attack rate of pupils to staff was 1% in one study.[30]

Regarding transmission among school staff, values of 1.29%,[38] 3.5% [29] and 4.4% [30]and 16.6% [32] were reported. Two studies showed no transmission among staff.[39, 40]

Risk of infection in seroprevalence studies

Four studies [25, 26, 28, 37] describe the detection of antibodies in school contexts in Germany, France and the USA. The detail information is provided in supplementary table 3.

In Germany, analysis of 13 schools in Saxony showed past infection in 0.2% of teaching staff and 0.7% of students, with an average of 0.6%.[25] In comparison, seroprevalence in northern France was 25.9% on average, with 28.75% in teaching staff and 12.8% in students.[28] In the USA, 14 days after a school index case, 1.66% of students and 0% of teachers tested positive for antibodies.[26] In the Midwest of the USA, 1.7% of teaching staff had a history of infection.[37]

Stratification of studies according to risk of infection after index case during data collection period

In order to better classify these heterogeneous results, study results were differentiated by two aspects: firstly, into three categories according to the prevalence of infection at the time of data collection and, secondly, according to the prevailing measures in schools at that time (Table 3). The detailed information about stratification of studies according to infection dynamics and NPIs during data collection period is provided in supplementary table 4.

Three studies conducted while the incidence of infection was low found no secondary cases following index cases.[27, 35, 42] Five studies conducted while the incidence of infection was in the medium range, reported that 0 to 4.4% of school staff and 0 to 6.5% of students developed secondary infections following index cases.[29, 30, 36, 38, 40] Two studies conducted while population infection incidence was

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reported that up to 16% of school staff developed infections following index cases, and up to 13% of students.[32, 39]

Regarding seroprevalence studies, two studies show an infection risk of 0 and 0.2% for teachers in a medium incidence of infection environment [25, 26] whereas two studies conducted at a higher incidence of infection showed a seroprevalence of 1.7 and 28%.[28, 37]

Table 3 Studies on the risk of new infections after index case in schools (attack rate) and the risk of infection based on seroprevalence in students and teachers

	Schools open with/without NPIs at the time of data collection				**Schools partially or completely closed at the time of data collection**			
	Infection risk of students	Attack rate students	**Infection risk teachers**	**Attack rate teachers**	**Infection risk students**	**Attack rate students**	**Infection risk teachers**	**Attack rate teachers**
Low infection incidence at the time of data collection **Case peak 0-10** **Death Peak <1**	No studies	No infections after initial infection (1 study)	No studies	No infections after initial infection (1 study)	No studies	No infections after initial infection (2 studies)	No studies	No infections after initial infection (2 studies)
Medium incidence infections at the time of data collection **Case peak 10-150** **Death peak < 5**	No studies	0-6.5%, higher in secondary school (5 studies)	No studies	0-4.4%, higher among school staff than pupils (5 studies	0.7-1.7% (seroprevalence, 2 studies)	No studies	0-0,2% (seroprevalence, 2 studies)	No studies
High incidence of infection at the time of data collection **Case peak > 150** **Death peak >5**	No studies	0.1-13% (2 studies)	No studies	0-16.6% (2 studies)	12.8% (1 study)	No studies	1.7-28% (2 studies)	No studies
Low incidence of infection: per 1 million population: peak daily case rate less than 10/day, peak case rate<1/day Mean incidence of infection: per 1 million population: peak number of daily cases below 150/day, peak number of deaths<5/day High incidence of infection: per 1 million population: peak daily case rate over 150/day, peak fatality rate>5/day								

Comparison of the risk of infection of teachers and other population groups

Two studies[31, 33] describe the risk of infection in Sweden. Here, during a period of high infection incidence, secondary schools were closed and pupils were taught in distance, while primary schools remained open and face-to-face teaching continued. The relative risk (RR) and 95% confidence interval (CI) for teachers in open primary schools was 1.1 (0.9-1.3), whereas RR and 95% CI for teachers in closed schools was 0.7 (0.5-1).[31]

The chance for primary school teachers to become infected with SARS-CoV-2 was about twice as high as that of secondary school teachers in distance, with the odds ratio (OR) and 95% CI of 2.01 (1.52-2.67). Partners of primary school teachers and parents of primary school students also had an increased chance of becoming infected, OR 1.3 (1-1.68) and OR 1.15 (1.03-1.27), when compared to secondary schools cohorts.[33] The comparison of infection risks and attack rates of school staff with other population groups is presented in table 4.

Table 4 Comparison of infection risks and attack rates of school staff with other population groups

	Schools open with / without NPIs			**School (partially) closed**		
Infection dynamics	Comparison students/teachers	Comparison teachers/teachers	Comparison teachers/population	Comparison students/teachers	Comparison teachers/teachers	Comparison teachers/population
low Case peak 0-10 Death peak <1	**Attack rates:** ***similar***, no RR calculable (1 study)	No studies	No studies	**Attack rates:** ***similar*** (2 studies)	No studies	No studies
Medium Case peak 10-150 Death peak 0,5-5	**Attack rate**: **Higher** in teachers (RR 1.6-4.4, 3 studies) **Lower** in teachers (RR n.c., 2 studies) **Same** (1 study)	No studies	No studies	**Infection risk** ***lower*** in teachers (RR=0.3,1 study**)**	No studies	No studies
High Case peak 90-1000 Death peak 5-20	**Attack rate:** ***Higher*** in teachers (RR 1.2 1 study) ***Lower*** in teachers (NR, 1 study)	No studies	**Infection risk** After school opening higher (1.42, 1 study) **Hospitalisation** After school opening similar (0.97, 1 study)	No studies	**Infection risk:** ***Same to higher*** in teachers in ***presence*** compared to distance (1.1. – 2.0, 2 studies)	**Infection risk:** Before school opening lower (RR 0.5, 1 study) **Hospitalisation**: Before school opening lower (RR 0.5, 1 study)

A recent study from Scotland compares the risk of infection as well as the risk of hospitalisation of teachers during a period of high infection incidence with school closures and a period of lower infection incidence and open schools with both hospital staff and the general population. The risk of infection as well as the risk of hospitalisation of teachers during school closures is about half that of the general population (RR 0.5). Following school openings, the risk of infection increased threefold and is higher than that of the general population (RR 1.42) and the risk of hospitalisation doubles and is similar to that of the general population (RR 0.97).[41]

DISCUSSION

Upon stratification of heterogeneous results in this review of infection risk and secondary attack rates of SARS-CoV-2 infection in school staff, we show that with a low incidence of infection attack rates are rather low and similar among teachers and students. When incidence and mortality ratios of SARS-CoV-2 were in the medium range, secondary attack rates in school were higher and higher for teachers than among students (0 to 6.6%). In settings with high infection dynamics during data collection (incidence >25/ 7 days/100,000, deaths per day >5/million population) intervals, the risk of infection following outbreaks in schools is usually higher among teachers than among students (up to 16%),[32] and the risk of infection via seroprevalence studies is up to 28%.[28]

Infectious students tend to infect other students rather than teachers. The student to staff transmission rate is low, i.e. 0%, compared to staff to student transmission, which was 1% in the same setting.[30] This is in line with several studies suggesting low secondary transmission from students to teachers in different countries.[12, 26] Infectious teachers tend to infect other teachers rather than students.[32, 36] This is supported by a study from Australia[44] and other transmission studies.[29, 30, 45] The study summarises that in the school setting the transmission risk is higher among adults and infectious children are less likely to infect teachers.[44]

During intervals of high population infection incidence the risk of infection increased among teachers in face-to-face classes compared to teachers in distance classes (RR 1.1-2.0 risk of infection)[31] and the risk of infection as well as the risk of hospitalization increased among teachers during school openings compared to school closings (1 study, RR=3 for infection risk and 1 study, RR=2, for hospitalization risk).[41]

Compared to the general population, the risk of infection and hospitalisation was lower for teachers during school closures than for the general population (RR=0.5 in one study) and increased (RR=1.42) after re-opening compared to the general population, while hospitalisation risk was not increased (RR=0.97) concordantly[41]. Thus, continuous presence of teaching staff in schools compared to intervals of or teachers in distance learning increases the risk of infection and also hospitalisation

This highlights the importance of transmission control measures such as contact tracing and fast quarantine orders. Upon detection of a single or few infections in schools, quarantine and testing strategies can help to prevent larger outbreaks.[46] During large outbreaks transmission directions are less defined and attack rates are much higher.[32]

However, the dependence on local arrangements and testing strategies of the evidence presented is critical. For example, if only symptomatic cases are tested or only reported cases are evaluated, this can lead to high numbers of unreported asymptomatically infected or untested infected people. This

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distorts the comparison between teachers and pupils, as children are more likely to become asymptomatically ill.

Similarly, seroprevalence studies reveal a heterogeneous picture with low evidence of infection incidence in the example of schools in Saxony, Germany during a data collection period with medium infection dynamics. However, the formation of antibodies is dependent on the intensity of the infection and immune response and can thus be underestimated, especially for children. Besides, it is difficult to reconstruct whether all detected infections occurred in the school environment.

Despite of heterogeneity in the included studies, two conclusions can be drawn from this review. Firstly, documenting local infection dynamics and implemented NPIs during data collection periods is crucial to understanding transmission dynamics in schools. Not all studies report these consistently. During periods of low incidence in the local population and schools with NPIs in place the risk to school staff is not necessarily higher than that of the general population and not comparable to the risk related to other high-risk professions such as health care staff. Studies reporting periods of high incidence are scarce but do show higher risk to school staff in these situations during periods where schools are not closed or NPIs are only partly in place. This may be due to the higher number and proximity of daily contacts in open schools compared to a general population under NPI public health measures.

Secondly, implementing screening and testing in schools is essential. In most of the included studies children seem less susceptible to SARS-CoV-2 infection. Students are less likely to transmit the virus to their peers or to teachers in the school setting. A large meta-analysis of prevalence studies [3] and school outbreak studies [47] supports this finding. However, these findings are biased by test strategies. If only symptomatic persons are tested and children show less symptoms, the number of positive cases in children is underestimated. Mass screenings of asymptomatic populations decrease the transmission of SARS-CoV-2.[48] Mass testing after index cases and frequent testing of asymptomatic students and staff was shown to reduce transmission in schools, albeit not specifically the infection risk of staff.[49] Mass testing and serial contact tracing and testing coupled with isolation and physical distancing can reduce the transmission SARS-CoV-2 in schools.[50, 51]

In Germany, schools were reopened in February 2021 despite rising population incidences (predominantly due to increased endemicity of the variant B1.1.7, now accounting for over 70% of cases in Germany). [52] Already a rise in cases among school-aged children is reported by the Robert Koch Institute and the national average incidence exceeds 100 cases / 100000 / 7 days [52]. Applying the conclusions to this scenario, we expect an increasing risk to school staff and students as social contacts in open schools will outnumber out-of-school contacts in a high community NPI and infection

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environment. Whereas the political discourse focuses primarily on the contribution of school cases to the overall infection dynamics, the reverse dependence of the infection risks in schools on community incidences and the associated health risk to staff and students is less discussed. Presumably, the school population is misleadingly thought of as young students (only) and thus considered to be less at risk of adverse outcomes. As we have demonstrated, the staff population has to be somewhat separated from the student population in terms of infection and transmission risks. Consequently, the risk to teachers and household contacts of students and staff should be considered more prominently in the balancing political decision around school openings and closures.

With that in mind, we recommend that legislators implement well-designed mass testing and serial contact tracing and testing strategies, also including asymptomatic individuals, to minimize the risk of school outbreaks during high infection dynamics

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Legends:

- Figure 1: PRISMA flow diagram
- Table 1: PICOs for included studies
- Table 2: Characteristics of included studies
- Table 3: Studies on the risk of new infections after index case in schools (attack rate) and the risk of infection based on seroprevalence in students and teachers
- Table 4: Comparison of infection risks and attack rates of school staff with other population groups
- Supplementary table 1: Risk of bias assessment of included studies
- Supplementary table 2: Secondary transmission or attack rates in school settings
- Supplementary table 3: Infection risk in school settings
- Supplementary table 4: Infection risks and attack rates according to infection dynamics and NPIs during data collection period

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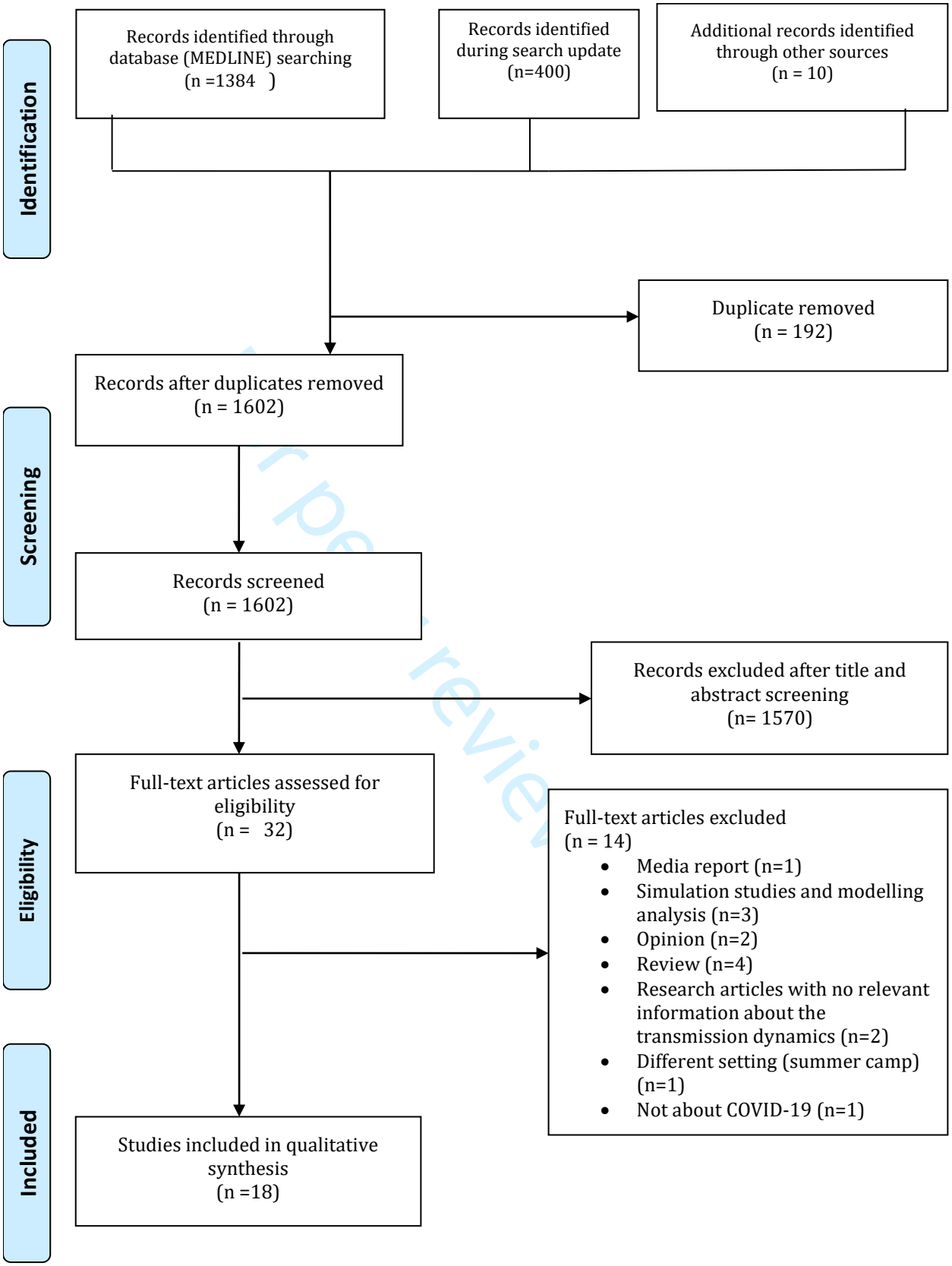


Figure 1: PRISMA (Preferred Reporting Items for Systematic Review and Meta-Analysis) flow diagram

Supplementary Table 1 Risk of Bias assessment of included studies

First author	Selection Bias	Performance Bias	Attrition Bias	Detection Bias	Reporting Bias	Information Bias	Overall
Macartney, K [30]	Medium	Low	Medium	Low	Low	Low	**Medium**
Stein-Zamir,C [32]	Low	Low	Medium	Low	Low	Low	**Medium**
Larosa, E [40]	Low	Low	Medium	Low	Low	NA	**Medium**
Public Health Agency Sweden [31]	Low	Low	Unclear	Low	Low	Unclear	**Unclear**
Vlachos, J [33]	Low	Low	Low	Low	Low	Low	**Low**
Brown, NE [26]	Medium	Unclear	High	Medium	High	High	**High**
Hernandez, A [29]	Low	Low	Medium	Low	Low	Low	**Medium**
Lopez, L [37]	Low	Low	Low	Low	Low	Low	**Low**
Heavey; L [42]	Medium	Low	Low	Low	Medium	Low	**Medium**
Yung, CF [34]	Low	Low	Low	Low	Low	Low	**Low**
Armann, JP [25]	Low	Low	Low	Low	Low	Low	**Low**
Fontanet, A [28]	Low	Low	Low	Low	Low	Low	**Low**
Danis, K [27]	Low	Medium	Low	Low	Low	Medium	**Medium**
NCIRS [35]	Medium	Low	Medium	Low	Low	Low	**Medium**
NCIRS [38]	Medium	Low	Medium	Low	Low	Low	**Medium**
Ismail, SA [36]	Medium	Low	Low	Low	Low	Low	**Medium**
Falk,A [39]	Medium	Low	Low	Low	Low	Low	**Medium**
Lynda, F [41]	Low	Low	Low	Low	Low	Low	**Low**

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Supplementary Table 2 Secondary transmission or attack rates in School setting

Source	First author	Main issue	Study type	Population(N)	N primary/index cases	N primary cases staff	N primary cases student	N Contact staff	N contacts student	N contacts	N contact tested	Setting	Comparator	Date of data collection	Outcome	N tested positive	Overall AR	AR teacher	AR Student	Comments	Infection dynamics[1]	Incident cases and deaths per day[2]	Control measures
Attack Rate in School																							
Lancet Child and Adolescent Health	Macartney , K [30]	transmission of COVID-19 among children and staff in schools and early childhood education and care (ECEC) setting	Perspective cohort study	1475(27+1448)	27 (12 children and 15 adults)	15	12	NA	NA	1448	633	15 Schools and 10 ECEC in New South Wales Australia	students	January 25-1 May 2020	Secondary attack rate	**18**	**1.2 %**	**child to staff = 1%** **staff to staff = 4.4%**	**child to child = 0.3%**	**18 secondary** infections: 10 infections were from children and 8 infection were from adults.	**265.65** cases per million **3.61** deaths per million	**Incidence:** Start: 0.16 case/ M Peak= 19.49 cases/ M End: 0.47 cases/ M **Deaths** Start: 0.04 Peak= 0.31 End= 0	Physical distancing, hygiene measures and educational facility cleaning Online classes from 23rd March.
Euro Surveillance	Stein-Zamir , C [32]	COVID-19 outbreak in School in Israel. The school has grade 7-grade 12.	Observational	1314 (2+1312)	2	NA	2	151	1161	1312	1312	SARS-CoV-2 **outbreak** in school of Israel after school reopening.	Students	May-June 2020	attack rate	**178 (Student= 153 and Staff= 25)**	**13.5 %**	**16.6 %**	**13.2%**	The number of cases was higher in lower grades (grade 7-9)	1089.61 cases per million 11.32 death per million	**Incidence** Start =13.40 Peak= 90.81 End= 90.81 **Deaths** Start= 0.46 Peak= 0.69 End= 0.12	Hygiene, face masks, social distancing and minimal interaction between classes.

[1] Total number of confirmed COVID-19 cases and deaths from COVID-19 during the data collection/ study duration in the country of study
[2] The number of incidence case per million and number of death per million per day. The data presented are at the start of the study period, peak during the study period and at the end of the study period.

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Euro Surveillance	Larosa, E [40]	Secondary transmission of COVID-19 cases in school and preschool setting of northern Italy	Observational	1248 (48+1200)	48	5	43	204	996	1200	1198 (994 student & 204 staff)	Total 36 different educational setting (school, preschool and infant toddler center) in northern Italy	Students of different school level	Sep 1-Oct 15	attack rate	**38**	**3.2 %**	**0**	**Overall = 3.8% Secondary School AR= 6.6% Elementary school AR= 0.38% Preschool AR= 0**	All the secondary cases were among the students who were tested	1842.70 cases per million 14.75 deaths per million	**Incidence** Start= 16.13 Peak= 145.6 End= 145.60 **Deaths** Start= 0.13 Peak= 1.37 End= 1.37	Physical distancing measures, mandatory wearing of surgical masks for children, single desk and desk must me 1 m apart, extracurricular activities were suspended.
Frontiers of Public Health	Hernandez, A [29]	Secondary transmission from the school staff	Case report	202(2+200)	2	2	NA	NA	NA	200 approximately	20 (10% of the contact)	School of Panama	Students	February-March 2020	attack rate	**7**	**3.5 %**	**3.5%**	**0**	All 15 close contacts of teachers tested negative for SARS-CoV-2 **Positive cases were the school contacts**	273.48 cases per million 6.72 deaths per million	**Incidence** Start= 0.23 Peak= 44.50 End= 44.50 **Death** Start= 0.23 Peak= 1.62 End = 1.39	School closure from March
Euro Surveillance	Heavey, L [42]	Secondary transmission in school setting	Observational	1031 (6+1025)	6	3	3	NA	NA	1025	NA	School in Ireland	NA	12 March 2020	Secondary transmission, SAR	**0 in School** **2 outside school**	**0**	**0**	**0**	**2 contacts of adult cases outside of school developed SARS-CoV-2 infection**	8.71 cases million 0.2 deaths per million	No cases and no deaths on the particular day in Ireland	School closure from 12th March

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																			with overall AR 0.17%			
Clinical Infectious Diseases	Yung, CF [34]	Secondary transmission in school setting	Observational	122 (3+119)	3	1	2	NA	NA	119	119 (8+34+77)	3 different school in Singapore	NA	Feb-March 2020	Secondary transmission, AR	**0**	**0**	**0**	**0**	No secondary cases were found among close contacts.	**155.55 cases per million** **0.17 deaths per million**	**Incidence** Start= 0.51 Peak= 12.48 End= 8.03 **Deaths** Start= 0.34 Peak= 0.34 End= 0	Terminal cleaning of school, suspension of extracurricular activities, preschool was closed for 14 days
Clinical Infectious Diseases	Danis, K [27]	Tertiary transmission in school setting	Observational	113 (1+112)	1	NA	1	NA	NA	112	101	3 schools and 1 ski club in French Alps	Student and teachers	Feb 2020	Tertiary transmission	**0**	**0**	**0**	**0**	No transmission in schools	1.44 cases per million 0.03 deaths per million	**Incidence** Start= 0.02 Peak= 0.31 End= 0.29 **Deaths** Start= 0.02 Peak= 0.02 End= 0	**Use of surgical masks**
NA	NCIRS [35]	secondary transmission in differed educational setting	Observational	527 (6+521)	6	2	4	62	459	521	319	6 different educational (5 schools and 1 ECEC)	Student; teacher	10 April – 3 July	Secondary transmission	**0**	**0**	**0**	**0**	No secondary transmissions	80.20 cases per million 1.96 deaths per million	**Incidence** Start= 4.20 Peak= 7.61 End=7.61 **Deaths** Start= 0.12 Peak= 0.31 End= 0	**Schools** were re-opened, full face-to-face teaching commenced while implementing COVID-19 safe practice

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NA	NCIRS [38]	secondary transmission in differed educational setting	Observational	3863(33+3824)	39	7	32	385	3439	3824	3641	34 different educational setting in NSW which includes 28 schools and 6 ECEC	student	July 4 – Sep 25	Secondary transmission; attack rate	**33**	**0.86 %**	**1.29 %**	**0.81%**	Among tested: AR= 0.9% High School AR= 1.1% Primary school AR= 0.4% ECEC AR= 0.7%	728.36 cases per million 30.04 deaths per million 744 cases across NSW. Among them 555 were locally acquired.	**Incidence** Start= 7.18 Peak= 28.08 End= 0.63 **Death** Start= 0 Peak= 2.31 End= 0.04	Schools were open for face-to-face learning while implementing COVID-19 safe practice.
Lancet Infectious Diseases	Ismail, SA [36]	estimating the rate of SARS-CoV-2 infection and outbreaks among staff and students in educational settings during the summer half-term	Prospective cross sectional analysis	928000 (Students only)	55	34	21	NA	NA	NA	NA	Schools in England	Student	June/July 2020	Secondary transmission	**NA**	**122**	**100**	**22**	Considering only outbreak situation there were 122 secondary cases in school during this period. Among them 100 cases were staff and 22 were students	661.80 cases per million 53.63 deaths per million	**Incidence** Start= 21.27 Peak= 21.65 End= 13.01 **Death** Start= 1.27 Peak= 3.80 End= 0.29	School closed from March 20 and reopened from June 1st after implementing smaller classes separated into distant social bubbles
MMWR	Falk, A [39]	Secondary transmission in	Observational	5530 (184+5346)	184	58	126	596	4750	5346	5530	Schools in Wisconsin, USA	Student	Aug-Nov 29, 2020	Secondary transmission;	**191**	**0.13 %**	**0**	**0.14%**	Among 191 tested positive, 7 were	26,802.94 cases per million	**Incidence** Start= 169.71	Compulsory masks for student and staff.

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		school setting													attack rate					secondary cases There was no secondary transmission among staff	339.08 deaths per million	Peak= 628.19 End= 423.22 **Deaths** Start= 3.31 Peak= 6.86 End= 3.13	Classroom size 11-20 students and indoor lunch.

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Supplementary Table 3 Infection risk in School setting

Source	First author	Main issue	Study type	Population (N)	N primary/ index cases	N primary cases staff	N primary cases student	N Contact staff	N contacts student	N contacts	N tested	Setting	Comparator	Date	Outcome	N tested positive	Overall risk of infection	Infection risk in teacher	Infection risk in student	Comments	Infection dynamics[1]	Incident cases and deaths per day[2]	Control measures
Infection risk in School																							
Public Health Agency Sweden	Public Health Agency Sweden [31]	Risk of SARS-CoV-2 infection in teachers in Sweden	Report (observational study)	Teachers working in different school in Sweden	364760	479 cases	NA	NA	NA	NA	NA	School of Sweden	Teachers of different school forms	March-May 2020	Relative risk	**NA**		**RR and 95%CI** High School teachers **0.7(0.5-1)** Primary school teachers **1.1(0.9-1.3)** Leisure educators **0.8(0.5-1.3)** Preschool teachers **0.7(0.6-0.9)** Other educators with theoretical specialists	NA	The relative risk is higher for primary school teachers and student assistants compared to other groups working in schools in Sweden	3,800.47 cases per million 435.08 deaths per million	**Incidence** Start= 0 Peak= 80.01 End= 26.24 **Death** Start= 0.1 Peak= 18.32 End= 0	Partial school closure

[1] Total number of COVID-19 cases and deaths from COVID-19 per million during the period of data collection/study duration in the country of study
[2] The number of incidence case per million and number of death per million per day. The data presented are at the start of the study period, peak during the study period and at the end of the study period

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																		1(0.7-1.5) Nanny **1(0.8-1.2)** Student assistants and other **1.1(0.8-1.4)**					
medRvix	Vlachos,J [33]	Risk of COVID-19 infection in the parents, high school teachers, lower secondary school teacher and their partners	Retrospective data analysis	Upper secondary and lower secondary teachers, their partners and student's parents from Sweden	NA	NA	NA	NA	NA	NA	NA	School and household of Sweden	Teachers of different school form, their partners And parents of students	June 15	Odds ratio and 95%CI	**NA**	**lower secondary school** **parents** **1.15 (1.03-1.27)** **Partners** **1.30(1.00-1.68)**	**2.01(1.52 – 2.67)**	NA	Secondary school is the reference group	2,833.37 cases per million 92.09 deaths per million	**Incidence** Start= 64.16 Peak= 168.13 End= 79.19 **Deaths** Start= 0.79 Peak= 10.69 End= 5.25	Partial school closure
Emerging Infectious Diseases	Brown, NE[26]	Teacher to student transmission	Observational (Serological survey)	121 (1+120)	1	1	NA	NA	120	120	21	School in USA.	Students	Blood sample collection date: March 13, 2020	Infection rate	**2**	**1.66%**	**0**	**1.66%**	blood collection was done 14 days after exposure	587.87 cases per million 16.13 deaths per	**Incidence** Start= 0.02 Peak= 79.50 End= 79.50 **Death**	NA

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																				million	Start= 0 Peak= 3.31 End= 3.31	
medRvix	Lopez, L [37]	Sero-prevalence of anti-SARS-CoV-2 IgG antibodies in school staff in Midwestern USA	Population based sero prevalence study	1261 eligible School staff	NA	NA	NA	NA	NA	NA	753 participated in the survey (Annual check-up)	18 year or older staff of Lake Central School Corporation located in suburban Indiana	General population	July 2020	Seroprevalence of antibody	22 participants tested positive for SARS-CoV-2 antibodies	2.9% (95%CI 1.8-4.4%) **When Corrected for performance parameter** 1.7%	1.7%		The seroprevalence of LCSC staff is 1.7% which is below the seroprevalence reported in metropolitan areas throughout the USA	5,658.69 cases per million 78.18 deaths per million	**Incidence** Start= 156.52 Peak= 229.03 End= 207.54 **Death** Start= 2.14 Peak = 4.33 End= 3.72	School closure, While entering the building for data collection facemask, social distancing and hand sanitizer were mandatory
medRvix	Armann, JP [25]	Seroprevalence study in student and teacher to assess their role SARS-CoV-2 transmission	Seroprevalence study	2045 (1538 students and 507 teachers)	NA	NA	NA	NA	NA	NA	2045	Secondary school in Saxony, Germany	students	25th may and 30th June	Prevalence (Seroprevalence)	12 (11 students and 1 teacher)	0.6%	0.2%	0.7%	Seroprevalence study in 13 different secondary school in Saxony, Germany	374.07 cases per million 26.90 deaths per million	**Incidence** Start= 12.75 Peak= 15.13 End= 4.49 **Death** Start= 1.35	School closure, face mask and social distancing

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																					Peak= 3.37 End= 0.17	
medRvix	Fontanet, A [28]	to assess the infection attack rate using serological assays	Retrospective cohort study (Seroprevalence study)	661 (including 240 pupil, 211 parents, 127 sibling, 53 teacher; 27 non-teaching staff and 3 other)	NA	NA	NA	NA	NA	NA	661	Secondary school and household	Student, parents	30 March -4 April	Infection attack rate	171 (pupil= 92; teacher= 23; school staff= 16; parents=24; sibling = 13, and other= 3)	25.9% (22.6-29.4)	12.18%	28.75%	Seroprevalence study in secondary school and household in Oise in Northern France	41.81 cases per million 69.49 deaths per million	**Incidence** Start= 66.70 Peak= 116.88 End=13.10 **Death** Start= 6.40 Peak= 19.18 End= 16.13	NA
medRvix	Lynda, Fenton [41]	Risk of COVID-19 and risk of hospitalization with COVID-19 among teachers compared with healthcare workers and general population of working age	Case control study	871568	NA	NA	NA	NA	NA	NA	NA	School, hospital and household of Scotland	Healthcare worker, Household member of healthcare worker and teacher	March 2020-January 2021	Risk rate/ odds ratio	NA		**Before re-opening**: Any cases 0.48 (0.39-0.60); Hospitalization 0.51 (0.31-0.82) **School re-opened:** Any cases 1.42	**Before reopening,** **Any case** HH_Teacher 0.78 (0.66-0.91) HCW 8.75 (8.29-9.23); HH_HCW 2.32 (2.11-2.56) **Hospitalization** HH_Teacher 0.63	comparators are healthcare workers and household member of teachers and healthcare worker The results presented are adjusted ratios obtained	56389.96 cases per million 1566.83 deaths per million	**Incidence** Start= 0.03 Peak= 1004.51 End= 312.14 **Death** Start= 0.01 Peak= 26.90 End= 17.75	School close and again reopen

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																		(1.35-1.49); Hospitalization 0.97 (0.72-1.29)	(0.44-0.91); HCW 3.15 (2.71-3.66); HH_HCW 1.87 (1.52-2.30) **Re-opened** **Any cases** HH_Teacher 1.01 (0.95-1.06); HCW 2.41 (2.33-2.49); HH_HCW 1.25 (1.20-1.31) **Hospitalization** HH_Teacher 0.97 (0.74-1.27); HCW 1.82 (1.55-2.14), HH_HCW 1.04 (0.83-1.23)	from logistic regression			

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Supplementary Table 4 Infection risks and attack rates according to infection dynamics and NPIs during data collection period

Infection dynamics cases	Infection dynamics deaths	control measures	Study	Infection risks for students	Attack rates for students	Infection risks for teachers	Attack rate for teachers	Comparator of infection risk for students/teachers	Comparator for teachers/teachers	Comment
Total number of cases in country / million / week < 100	N of deaths per million in data collection period < 100	School open	Heavey, L [42]		0		0			No secondary transmission in school setting
		School open with NPIs	Danis, K [27]		0		0			No secondary transmission in school setting
			NCIRS (2nd term) [35]		0		0			No secondary transmission in school setting
		NA	Fontanet, A [28]	12.18 %		28.75%				
Total number of cases in country / million / week > 101 and < 1000	N of deaths per million in data collection period < 100	School open	Hernandez, A [29]		0		3.5 %			
			Yung, CF [34]		0		0			No secondary transmission
		School open with NPIs	Macartney, K [30]		0.3 %			Child to staff transmission rate: 1%	Staff to staff transmission rate 4.4%	
		School close/ online teaching	Armann, JP [25]	0.7 %		0.2%				
		School open	NCIRS (3rd term) [38]		0.81%		1.29%			

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		with NPIs	Larosa, E [40]		3.8 %		0	Secondary school= 6.6%		
								Elementary school = 0.38%		
		School open with NPIs	Ismail, SA [36]		N= 22		N= 100			Absolute number of secondary infections during outbreak in school
		NA	Brown, NE [26]	1.66%		0				
Total number of cases in country / million / week > 101 and < 1000	N of deaths per million in data collection period > 50 and <100	School open with NPIs	Falk, A [39]		0.14%		0			
			Stain-Zamir, C [32]		13.2%		16.6%			
		Partial school closure	Public health Agency, Sweden [31]						High school= 0.7(0.5-1)	Relative risk among teachers
									Primary school= 1.1(0.9-1.3)	
									Preschool= 0.7 (0.6-0.9)	
									Student assistant= 1(0.8-1.4)	
			Valchos, J [33]						OR=2.01 (1.52-2.67)	Secondary school teachers are a reference group

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		School closes/ online	Lynda, F [41]	NA	NA	0.48(0.39- 0.60)				
			Lopez, L [37]	0		1.7%				Seroprevalence assay done only in staff
		School reopen with NPI	Lynda, F [41]	NA	NA	1.42 (1.35 - 1.49)				

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Reporting checklist for systematic review and meta-analysis.

Based on the PRISMA guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

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	Reporting Item	Page Number
Title		
#1	Identify the report as a systematic review, meta-analysis, or both.	1
Abstract		

Structured summary	#2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number	2
Introduction			
Rationale	#3	Describe the rationale for the review in the context of what is already known.	4
Objectives	#4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	NA
Methods			
Protocol and registration	#5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address) and, if available, provide registration information including the registration number.	5
Eligibility criteria	#6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rational	5
Information sources	#7	Describe all information sources in the search (e.g., databases with dates of coverage, contact with study authors to identify additional studies) and date last searched.	5
Search	#8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5

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Study selection	#9	State the process for selecting studies (i.e., for screening, for determining eligibility, for inclusion in the systematic review, and, if applicable, for inclusion in the meta-analysis).	5-6
Data collection process	#10	Describe the method of data extraction from reports (e.g., piloted forms, independently by two reviewers) and any processes for obtaining and confirming data from investigators.	6
Data items	#11	List and define all variables for which data were sought (e.g., PICOS, funding sources), and any assumptions and simplifications made.	6
Risk of bias in individual studies	#12	Describe methods used for assessing risk of bias in individual studies (including specification of whether this was done at the study or outcome level, or both), and how this information is to be used in any data synthesis.	6
Summary measures	#13	State the principal summary measures (e.g., risk ratio, difference in means).	6
Planned methods of analyis	#14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I2) for each meta-analysis.	NA
Risk of bias across studies	#15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	NA
Additional analyses	#16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which	NA

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		were pre-specified.	
Results			
Study selection	#17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7
Study characteristics	#18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citation.	9-10
Risk of bias within studies	#19	Present data on risk of bias of each study and, if available, any outcome-level assessment (see Item 12).	7-8
Results of individual studies	#20	For all outcomes considered (benefits and harms), present, for each study: (a) simple summary data for each intervention group and (b) effect estimates and confidence intervals, ideally with a forest plot.	NA
Synthesis of results	#21	Present the main results of the review. If meta-analyses are done, include for each, confidence intervals and measures of consistency.	11-13
Risk of bias across studies	#22	Present results of any assessment of risk of bias across studies (see Item 15).	7-8
Additional analysis	#23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	NA
Discussion			

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Summary of Evidence	#24	Summarize the main findings, including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., health care providers, users, and policy makers	14-15
Limitations	#25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review level (e.g., incomplete retrieval of identified research, reporting bias).	14-15
Conclusions	#26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	15
Funding			
Funding	#27	Describe sources of funding or other support (e.g., supply of data) for the systematic review; role of funders for the systematic review.	17

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The risk of infection and contribution to transmission of SARS-CoV-2 in school staff - a systematic review

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ABSTRACT

Objective: To summarize the comparative risk of infection in school staff and their contribution to SARS-CoV-2 transmission.

Design: Systematic review using Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) guideline.

Data sources: MEDLINE, WHO COVID database and preView were searched on 29 January 2021.

Eligibility criteria: We included studies that reported risk of SARS-CoV-2 infection in school staff or transmission of SARS-CoV-2 in school settings.

Data extraction and synthesis: Data extraction was done in duplicates. Data synthesis was qualitative. We report attack rates and infection risk in school settings for staff and students stratified by control measures taken and infection dynamics at the point of data collection.

Results:

Eighteen studies were included. Three studies in low incidence settings showed low attack rates similar for teachers and students. Five studies in medium incidence settings and two studies in high incidence settings showed secondary attack rates up to 16% in school staff.

Seroprevalence studies, two in each low and high incidence settings showed an infection risk of 0-0.2% and 1.7-28% for teachers.

The risk of infection for teachers compared to students were similar in one study in low incidence setting, higher in three studies (RR 1.2-4.4) and lower in three studies in medium to high incidence settings. The risk of infection for teachers in a high infection environment is higher in face-to-face than in distance classes when compared with general population groups. The risk of infections as well as risk of hospitalization both increased for teachers during school openings compared to school closure.

Conclusion: While in low incidence settings there is little evidence for school staff to be at high risk of SARS-CoV-2 infection, in high incidence settings there is an increased risk of SARS-CoV-2 infection in school staff teaching face-to-face compared to staff teaching digitally and general population.

Key words: SARS-CoV-2, attack rate, infection risk, school staff, students

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Article Summary

Strength and Limitations of this study:

- The study results were stratified according to the prevalence of infection during data collection period and prevailing control measures in the school setting at that time.
- The infection risk in teachers/school staff were compared with infection risk in students, general population, and teachers.
- The results from included studies were heterogeneous.

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INTRODUCTION

Corona virus disease-19 (COVID-19) is a global public health threat, caused by severe acute respiratory syndrome coronavirus-2 (SARS-CoV-2).[1] Although people of all ages are affected, the severity of the clinical course increases with age (more severe in people >65 years of age).[2, 3] Children and adolescents most commonly experience a mild clinical course and show less severe outcomes compared to adults and ageing people.[4-7] When showing severe outcomes, long-term complications can be equal or worse in children than in adults.[8]

Non-pharmaceutical interventions (NPIs) like isolation, quarantine and social distancing including large-scale school closures are applied near-universally to curb the transmission of SARS-CoV-2.[9, 10] Such conventional public health measures appear to reduce the number of new infections.[10, 11] However, school closures alone are not sufficient to prevent community transmission of SARS-CoV-2.[12, 13]

Several systematic reviews, meta-analyses and large ecological analyses have focused on effects and adverse effects of school closures mainly assessing endpoints concerning the effect on community transmission as well as effects on children.[9, 14, 15] Long-term school closures are a threat to the physical and mental health of children and adolescents and intensify the racial and socioeconomic gaps in society.[16-20]

Nevertheless, keeping schools open when community transmissions are increasing may be posing a threat to school staff in particular, as their age leaves them more at risk of severe infections compared to students. Evaluating the risk to school staff as well their role in schools and community transmission is thus essential to an evidence-based approach to pandemic public health strategies.

In an umbrella review (Lange et al, submitted) we did not find any systematic review focusing on risk of and contribution to transmission of school staff.

The risk of infection in school staff in dynamic infection environments depends on the population infection dynamic as well as the infection dynamic within schools, the susceptibility of staff to the infection and the number of contacts of the staff at that time. An absolute estimate of the risk of infection is futile due to its dependence on the evolving context. We have therefore collated the existing evidence on the relative risk of infection compared to other population groups in original papers and existing reports and stratified by infection dynamic prevalent during the period of data collection.

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METHODS

Protocol and registration

We followed PRISMA guidelines[21] and registered this review with PROSPERO (CRD42021239225).

Search strategy

We searched MEDLINE and preView most recently on 29th January 2021 by using search terms "SARS-CoV-2"/ "COVID-19" and "teacher"/"school" combinations with OR and AND Boolean operators. We also searched WHO COVID-19 database for relevant literatures.

We did not restrict our search to any study design or language of publication. Preprints are included in this search if available from preprint databases. We did not consider the preprint available only on homepages or institute websites.

Eligibility criteria

Studies reporting the risk of COVID-19 in teachers or any school staff or any kind of involvement of teacher or school staff in SARS-CoV-2 transmission were eligible for inclusion in the review. Articles published in peer-reviewed journals, preprints, technical reports, and case reports were included. Studies and reports were also included based on expert suggestion.

Modelling studies, opinion analysis, media reports, reviews, and meta-analysis were excluded. We also excluded studies reporting SARS-CoV-2 transmission in students and school staff but in different school settings and studies reporting solely risk factors for SARS-CoV-2 infection in teachers or students. The Patients/Population, Intervention, Comparison and Outcomes (PICOs) for the included studies is presented in table 1.

Table 1 Patients/Population, Intervention, comparison, and Outcomes (PICOs) for included studies

Transmission of SARS-CoV-2 in school	Patients/Population	Intervention/ Exposure	Comparator	Outcome
	School staff, Any contacts of school staff	School, primary school, secondary school	School children, general population, present in the school (distance learning), school staff in different school forms or learning situations	Secondary attack rates as reported by authors of original papers, relative/infection risk, odds ratio,

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Study selection

Two reviewers (SJK and AJ) screened the title and abstracts and read the full-text independently based on the predefined eligibility criteria. Inconsistencies and disagreements in the judgement were resolved by consultation with a third reviewer (BL).

Patient and Public Involvement

Patients or the public were not involved in any stage of this systematic review.

Data Extraction

Two reviewers (SJK and AJ) independently extracted the data from included studies into a pre-specified form. Disagreements in the data extraction process were resolved by consultation with the third reviewer (BL). Data related to study characteristics (source, name of first author, study design/ type, date of data collection), study population (population of staff, population of students and population of contacts), main issue, study setting, comparator, attack rate in staff, attack rate in student, infection risk in student, infection risk in staff, outcome and results were extracted.

Quality appraisal

The Agency for Healthcare Research and Quality (AHRQ) checklist was adapted to assess the quality of included studies. [22-24]

The risk of bias domains used were selection bias, performance bias, attrition bias, detection bias, reporting bias and information bias. The overall risk of bias for included studies was classified as high, unclear, medium or low risk of bias. The criteria for *high risk of bias* for included studies are high risk of bias in any one of the domains. Studies with unclear risk of bias in any one of the key domains and no high risk of bias in any other domain were deemed *unclear risk of bias.* Studies with *medium risk of bias* in any one of the domains and low risk of bias in all other domains were deemed *medium risk of bias* and those with *low risk of bias* in all the domains were deemed *low risk of bias.*

Synthesis of results

Qualitative data synthesis was performed by describing study characteristics and main research questions, with the main conclusions of included studies presented narratively and in table format. The findings were presented based on the different type of SARS-CoV-2 transmission found in the school setting. When absolute numbers were available, we calculated (secondary) attack rates. When authors already calculated the attack rate we report them as given. Where infections risk is given by either seroprevalence or PCR based testing, we report them as given. All outcomes are reported stratified by infection environment and NPIs measures in place during data collection periods.

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RESULTS

Study selection and study characteristics:

The search yielded 1784 studies. Of these, eight met the inclusion criteria. A further 10 studies were found through screening references of systematic reviews, meta-analyses and following expert suggestions. Eighteen studies were included in the review; the selection process is described in Figure 1.

Almost all of the included studies were conducted in 2020. Ten of the included studies had a data collection/analysis period from January to June.[25-34] Three studies have collected data from April to July,[35] in June/July,[36] or in July only.[37] One study collected data from July to September [38] and two studies had data collection periods from August to November.[39, 40] One study uses data from March 2020 to January 2021,[41] and the remaining study analysed data from March 12, 2020.[42] During the data collection period, the total number of SARS-CoV-2 cases in the countries of study ranged from 1.44 cases/million to 26,802 cases/million and SARS-CoV-2-related deaths from 0.03 deaths/million to 339.68 deaths/million.[43] Similarly, the number of new cases per day at the start of the study interval ranges from 0 to 169.71 cases per million per day. At the end of the study period the incidence ranges from 0.29 to 423.22 cases per million per day.[43]

The studies were originated in Australia,[30, 35, 38] France,[27, 28] Germany,[25] Ireland,[42] Israel,[32] Italy,[40] Panama,[29] Singapore,[34] Sweden,[31, 33] the UK,[36] Scotland, [41] and the USA.[26, 37, 39] Among them three were reports published by the Public Health Agency of Sweden[31] and by the National Centre for Immunization Research and Surveillance (NCIRS), New South Wales, Australia.[35, 38] Five were published in preprint [25, 28, 33, 37, 41] with the remaining ten studies published in peer-review journals.

All included studies provide information about either risk of SARS-CoV-2 infection in teachers and/or students, transmission of SARS-CoV-2 in school settings or seroprevalence of SARS-CoV-2 IgG antibodies in school staff or school settings. Most of the included studies report attack rates,[27, 29, 30, 32, 34-36, 38-40, 42] seroprevalence of SARS-CoV-2 IgG [25, 26, 28, 37] or infection risk [31, 33, 41] among teachers and students. The characteristics of included studies are tabularized in table 2.

Quality Assessment:

Among the included studies, six have a low risk of bias, ten studies have a medium risk of bias, one has an unclear risk of and one has a high risk of bias. Reasons for assigning medium risk of bias were: not all contacts were tested in contact tracing studies and some of the studies used only case notification data from nationwide surveillance database. This increases the possibility of missing asymptomatic cases and cases that were not reported. The reason for assigning unclear risk of bias was that the occupation code was missing for 25% of the confirmed cases. It was unclear how the study group dealt

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with this issue. The reason for assigning high risk of bias to one of the studies was selective reporting of results, low participation rate and use of questionnaires to assess symptoms. This increases the possibility of recall and misclassification bias. The overall risk of bias assessments for included studies are tabularized (supplementary table 1).

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Table 2 Characteristics of Included studies

Source	First author	Region	Main issue	Type of study	Population	Population Staff	Population Contacts	Setting	Comparator	Date collection period	Outcome
Attack rate/secondary infections in school											
Lancet Child and Adolescent Health	Macartney; K [30]	Australia	Transmission of SARS-CoV-2 among children and staff in schools and early childhood education and care (ECEC) settings	Perspective cohort study	Total= 1475 27 primary infections (12 children and 15 adults) and 1448 contacts.	NA	1448	15 schools and 10 ECEC in New South Wales Australia 633 had nucleic acid or antibody testing or both	Students	25 January - 1 May 2020	Attack rate
Euro Surveillance	Stein-Zamir, C[32]	Israel	COVID-19 outbreak in school in Israel. The school hosts grades 7-12.	Observational	Total= 1314 2 primary infections among students. 1161 student contacts and 151 staff in the school	151	1312	SARS-CoV-2 outbreak in school in Israel after school reopening.	Students	May-June 2020	Attack rate
Euro Surveillance	Larosa, E [40]	Italy	Secondary transmission of COVID-19 cases in school and preschool setting in northern Italy	Observational	Total= 1248 48 index cases (43 students and 5 teachers) and 1200 contacts of primary cases.	209	1200	Total 36 schools and preschools in northern Italy	Students of different school levels	1 September – 15 October, 2020	Attack rate
Frontiers of Public Health	Hernandez , A[29]	Panama	Secondary transmission of school staff	Case report	Total= 202 2 primary cases (1 teacher and 1 director) Approx. 200 contacts in school Household contacts of primary cases	NA	200	School in Panama	Students	February-March 2020	Secondary attack rate, attack rate
Euro Surveillance	Heavey, L[42]	Ireland	Secondary transmission in school setting	Observational	**Total in school = 1031** Total including other setting = 1161 (6 cases= 3 paediatric and 3 adult)	NA	**In school=1031** (1155 contacts including other settings)	School in Ireland	NA	12 March 2020	Secondary attack rate, attack rate
Clinical Infectious Diseases	Yung, CF [34]	Singapore	Secondary transmission in school setting	Observational	Total = 122 (3 primary cases (1 student from secondary school, 1 student from preschool and 1 adult staff from other preschool)	NA	119 (8+34+77)	3 different schools in Singapore	NA	Febraury-March 2020	Secondary transmission, attack Rate
Clinical Infectious Diseases	Danis, K [27]	France	Tertiary transmission of SARS-CoV-2 in school setting	Observational	113 (1 secondary cases and 112 school contacts)	NA	112	3 different schools and 1 ski club in the French Alps	Teachers, students	February 2020	Tertiary transmission
National Centre for Immunisation Research and Surveillance, NSW, Australia	NCIRS [35]	Australia	Secondary transmission in different school settings	Observational	Total=527 6 primary cases (4 students and 2 staffs) and 521 close contacts from 6 different educational settings (5 schools and 1 ECEC)	62	521	6 different schools (5 schools and 1 ECEC) in NSW; Australia. The primary cases were found in the school setting and contact tracing was done and diagnostic test were performed in close contacts	Students	10 April -3 July, 2020	Secondary transmission, attack rate
National Centre for Immunisation Research and Surveillance, NSW, Australia	NCIRS [38]	Australia	Secondary transmission in different school settings	Observational	Total= 3857 39 primary cases (32 students and 7 staff) and 3824 close contacts in 34 different educational settings (28 schools and 6 ECEC)	385	3824	39 primary cases were found in 34 different educational settings (28 schools and 5 ECEC) in NSW Australia.	Students	4 July - 25 September, 2020	Secondary transmission, attack rate
Lancet Infectious Diseases	Ismail, SA [36]	UK	Estimating the rate of SARS-CoV-2 infection and outbreaks among staff and students in educational settings during the summer half-term (June–July, 2020) in England	Prospective cross sectional analysis	Students and staff in school of England	NA	NA	Schools in England	Students	June/July 2020	Secondary transmission
Morbidity and Mortality Weekly Report	Falk, A [39]	USA	Secondary transmission in the school setting	Observational	Total= 5530 184 primary cases with 5346 close contacts in 17 different educational settings in Wisconsin, USA. (4876 students and 654 staff in face-to-face learning)	654	5346	17 different schools settings; 8 elementary schools (k-6) and 9 secondary schools with grade 7 to 12 in Wisconsin, USA	Student	31 August – 29 November, 2020	Secondary transmission and attack rate

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Infection risk in Teachers											
Public Health Agency Sweden	Public Health Agency Sweden [31]	Sweden	Risk of SARS-CoV-2 infection in teachers in Sweden	Observational Report	364760 (age 44-53) Teachers working in different school in Sweden	NA	NA	Different school settings in Sweden	Teachers of different school levels	March-May 2020	Relative risk
medRvix	Vlachos, J[33]	Sweden	Risk of SARS-CoV-2 infection in parents, high school teachers, lower secondary school teacher and their partners were assessed after the partial school closure	Retrospective data analysis	Upper secondary and lower secondary school teachers, their partners and students' parents from Sweden	NA	NA	Upper secondary and lower secondary schools of Sweden	Teachers of different school levels and their partners. Parents of children attending different schools	15 June 2020	Odds ratio
medRvix	Lynda, F [41]	Scotland	Risk of COVID-19 and risk of hospitalization with COVID-19 among teachers compared with healthcare workers and general population of working age	Case control study	Total study population N= 871568 N teachers = 18479 N healthcare workers= 35461	18479	NA	Teachers and healthcare workers in Scotland	Health care workers, household members of healthcare workers and teachers	March 2020-January 2021	Risk rate
Seroprevalence studies											
Emerging Infectious Diseases	Brown,NE [26]	USA	Teacher to student transmission	Cross sectional (seroprevalence survey)	Total = 121 1 infected teacher and 120 student contacts	NA	120	School in the USA. All students were instructed to quarantine. 21 students participated in serological survey 14 days after quarantine	Students	Blood sample collection date: 13 March, 2020	Secondary infection
medRvix	Armann, JK [25]	Germany	Seroprevalence study in teachers and students	Cross sectional (Seroprevalence study)	2045 participants (1538 students and 507 teachers)	507	NA	13 different secondary schools in Saxony, Germany	Students	Blood sample collection 25 May and 30 June 2020	Seroprevalence
medRvix	Lopez, L [37]	USA	Seroprevalence of anti-SARS-CoV-2 IgG antibodies in school staff in Midwestern USA	Population based seroprevalence study	1261 eligible school staff	753 staff members participated in the survey	NA	18 year or older staff of Lake Central School Corporation located in suburban Indiana employed in academic year 2018/19 or 2019/20 and with record pf annual wellness check	General population	July 2020	Seroprevalence of antibody
medRvix	Fontanet, A [28]	France	Assessing infection attack rate using serological assays	Observational cohort (Seroprevalence study)	661 (pupils, parents, siblings, including teachers and non-teaching staff)	661	NA	Secondary school and household in Oise department in Northern France	Students and general population	30 March- 4 April, 2020	Infection attack rate

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Findings

Attack rates in school staff and students

Eleven studies reported data on attack rates in schools.[27, 29, 30, 32, 34-36, 38-40, 42] The detail information is provided in supplementary table 2.

Four studies found no secondary transmission in schools following index cases.[27, 34, 35, 42] Of the remaining seven studies, six reported attack rates of 0 to 13% following outbreaks among students and attack rates of 0 to 16.6% following outbreaks among school staff.[29, 30, 32, 38-40] One of the studies reports 100 secondary cases in staff and 22 in students related to one outbreak.[36]

Secondary attack rates among pupils were 0.14%,[39] 0.3%,[30] 0.81%,[38] and 3.8%.[40] The latter study further differentiated between 6.6% in secondary schools and 0.38% in primary schools with no secondary transmission in pre-schools.[40] The secondary attack rate of pupils to staff was 1% in one study.[30]

Regarding transmission among school staff, values of 1.29%,[38] 3.5% [29] and 4.4% [30]and 16.6% [32] were reported. Two studies showed no transmission among staff.[39, 40]

Risk of infection in seroprevalence studies

Four studies [25, 26, 28, 37] describe the detection of antibodies in school contexts in Germany, France and the USA. The detail information is provided in supplementary table 3.

In Germany, analysis of 13 schools in Saxony showed past infection in 0.2% of teaching staff and 0.7% of students, with an average of 0.6%.[25] In comparison, seroprevalence in northern France was 25.9% on average, with 28.75% in teaching staff and 12.8% in students.[28] In the USA, 14 days after a school index case, 1.66% of students and 0% of teachers tested positive for antibodies.[26] In the Midwest of the USA, 1.7% of teaching staff had a history of infection.[37]

Stratification of studies according to risk of infection after index case during data collection period

In order to better classify these heterogeneous results, study results were differentiated by two aspects: firstly, into three categories according to the prevalence of infection at the time of data collection and secondly, according to the prevailing measures in schools at that time (Table 3). The detailed information about stratification of studies according to infection dynamics and NPIs during data collection period is provided in supplementary table 4.

Three studies conducted while the incidence of infection was low found no secondary cases following index cases.[27, 35, 42] Five studies conducted while the incidence of infection was in the medium range, reported that 0 to 4.4% of school staff and 0 to 6.5% of students developed secondary infections following index cases.[29, 30, 36, 38, 40] Two studies conducted while population infection incidence was

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reported that up to 16% of school staff developed infections following index cases, and up to 13% of students.[32, 39]

Regarding seroprevalence studies, two studies conducted during a medium incidence of infection show an infection risk of 0 and 0.2% for teachers [25, 26] whereas two studies conducted during a higher incidence of infection showed a seroprevalence of 1.7 and 28%.[28, 37]

Table 3 Studies on the risk of new infections after index case in schools (attack rate) and the risk of infection based on seroprevalence in students and teachers

	Schools open with/without Non-pharmaceutical Interventions (NPIs) at the time of data collection				**Schools partially or completely closed at the time of data collection**			
	Infection risk of students	Attack rate students	**Infection risk teachers**	**Attack rate teachers**	**Infection risk students**	**Attack rate students**	**Infection risk teachers**	**Attack rate teachers**
Low infection incidence at the time of data collection **Case peak 0-10** **Death Peak <1**	No studies	No infections after initial infection (1 study)	No studies	No infections after initial infection (1 study)	No studies	No infections after initial infection (2 studies)	No studies	No infections after initial infection (2 studies)
Medium incidence infections at the time of data collection **Case peak 10-150** **Death peak < 5**	No studies	0-6.5%, higher in secondary school (5 studies)	No studies	0-4.4%, higher among school staff than pupils (5 studies	0.7-1.7% (seroprevalence, 2 studies)	No studies	0-0.2% (seroprevalence, 2 studies)	No studies
High incidence of infection at the time of data collection **Case peak > 150** **Death peak >5**	No studies	0.1-13% (2 studies)	No studies	0-16.6% (2 studies)	12.8% (1 study)	No studies	1.7-28% (2 studies)	No studies
Low incidence of infection: per 1 million population: peak number of daily cases less than 10/day, peak number of deaths <1/day Medium incidence of infection: per 1 million population: peak number of daily cases below 150/day, peak number of deaths<5/day High incidence of infection: per 1 million population: peak number of daily case over 150/day, peak number of deaths >5/day								

Comparison of the risk of infection of teachers and other population groups

Two studies[31, 33] describe the risk of infection in Sweden. Here, during a period of high infection incidence, secondary schools were closed and pupils were taught in distance, while primary schools remained open and face-to-face teaching continued. The relative risk (RR) and 95% confidence interval (CI) for teachers in open primary schools was 1.1 (0.9-1.3), whereas RR and 95% CI for teachers in closed schools was 0.7 (0.5-1).[31]

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The chance for primary school teachers to become infected with SARS-CoV-2 was about twice as high as that of secondary school teachers in distance, with the odds ratio (OR) and 95% CI of 2.01 (1.52-2.67). Partners of primary school teachers and parents of primary school students also had an increased chance of becoming infected, OR 1.3 (1-1.68) and OR 1.15 (1.03-1.27), when compared to secondary schools cohorts.[33] The comparison of infection risks and attack rates of school staff with other population groups is presented in table 4.

Table 4 Comparison of infection risks and attack rates of school staff with other population groups

	Schools open with / without Non pharmaceutical interventions (NPIs)			**School (partially) closed**		
Infection dynamics	Comparison students/teachers	Comparison teachers/teachers	Comparison teachers/population	Comparison students/teachers	Comparison teachers/teachers	Comparison teachers/population
low **Case peak 0-10** **Death peak <1**	**Attack rates:** ***similar***, no RR calculable (1 study)	No studies	No studies	**Attack rates:** ***similar*** (2 studies)	No studies	No studies
Medium **Case peak 10-150** **Death peak 0,5-5**	**Attack rate**: **Higher** in teachers (RR 1.6-4.4, 3 studies) **Lower** in teachers (RR n.c., 2 studies) **Same** (1 study)	No studies	No studies	**Infection risk** ***lower*** in teachers (RR=0.3,1 study)	No studies	No studies
High **Case peak 90-1000** **Death peak 5-20**	**Attack rate:** ***Higher*** in teachers (RR 1.2 1 study) ***Lower*** in teachers (NR, 1 study)	No studies	**Infection risk** After school opening higher (1.42, 1 study) **Hospitalisation** After school opening similar (0.97, 1 study)	No studies	**Infection risk:** ***Same to higher*** in teachers in ***presence*** compared to distance (1.1. – 2.0, 2 studies)	**Infection risk:** Before school opening lower (RR 0.5, 1 study) **Hospitalisation**: Before school opening lower (RR 0.5, 1 study)

A study from Scotland compares the risk of infection as well as the risk of hospitalisation of teachers during a period of high infection incidence with school closures and a period of lower infection incidence and open schools with both hospital staff and the general population. The risk of infection as well as the risk of hospitalisation of teachers during school closures is about half that of the general population (RR 0.5). Following school openings, the risk of infection increased threefold and is higher than that of the general population (RR 1.42) and the risk of hospitalisation doubles and is similar to that of the general population (RR 0.97).[41]

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DISCUSSION

Upon stratification of heterogeneous results in this review of infection risk and secondary attack rates of SARS-CoV-2 infection in school staff, we show that during a low incidence of infection at the time of data collection, attack rates are rather low and similar among teachers and students compared to medium and high incidence of infection. During a medium incidence and mortality rate of SARS-CoV-2 at the time of data collection, secondary attack rates in school were higher and higher for teachers than among students (0 to 6.6%). In settings with high infection dynamics during data collection (incidence >25/ 7 days/100,000, deaths per day >5/million population) intervals, the risk of infection following outbreaks in schools is usually higher among teachers than among students (up to 16%),[32] and the risk of infection via seroprevalence studies is up to 28%.[28]

Infectious students tend to infect other students rather than teachers. The student to staff transmission rate is low, i.e. 0%, compared to staff to student transmission, which was 1% in the same setting.[30] This is in line with several studies suggesting low secondary transmission from students to teachers in different countries.[12, 26] Infectious teachers tend to infect other teachers rather than students.[32, 36] This is supported by a study from Australia[44] and other transmission studies.[29, 30, 45] The study summarises that in the school setting the transmission risk is higher among adults and infectious children are less likely to infect teachers.[44]

In setting with high population infection incidence during data collection, the risk of infection was higher among teachers in face-to-face classes compared to teachers in distance classes (RR 1.1-2.0 risk of infection)[31] and the risk of infection as well as the risk of hospitalization increased among teachers during school openings compared to school closings (1 study, RR=3 for infection risk and 1 study, RR=2, for hospitalization risk).[41]

Compared to the general population, the risk of infection and hospitalisation was lower for teachers during school closures than for the general population (RR=0.5 in one study) and increased (RR=1.42) after re-opening compared to the general population, while hospitalisation risk was not increased (RR=0.97) concordantly[41]. Thus, continuous presence of teaching staff in schools compared to intervals of or teachers in distance learning increases the risk of infection and also hospitalisation

This highlights the importance of transmission control measures such as contact tracing and fast quarantine orders. Upon detection of a single or few infections in schools, quarantine and testing strategies can help to prevent larger outbreaks.[46] During large outbreaks transmission directions are less defined and attack rates are much higher.[32]

However, the dependence on local arrangements and testing strategies of the evidence presented is critical. For example, if only symptomatic cases are tested or only reported cases are evaluated, this

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can lead to high numbers of unreported asymptomatically infected or untested infected people. This distorts the comparison between teachers and pupils, as children experience a mild clinical course and fewer symptoms hence increasing the chance of being untested or not reported.

Similarly, seroprevalence studies reveal a heterogeneous picture with low evidence of infection incidence in the example of schools in Saxony, Germany during a data collection period with medium infection dynamics. However, the formation of antibodies is dependent on the intensity of the infection and immune response and can thus be underestimated, especially for children. Besides, it is difficult to reconstruct whether all detected infections occurred in the school environment.

Limitation of the Review

There are limitations of this review. First, we did not conduct quantitative meta-analysis since the heterogeneity among included studies make them less comparable and hence meta-analysis was not right choice in this situation. Second, the included studies did not explicitly mentioned whether they tested only symptomatic or reported cases or both symptomatic and asymptomatic cases. Testing mainly symptomatic cases might skew results towards higher infection risks in school staff as adults typically have a higher proportion symptomatic infections.[47] Third, we were not able to capture the endemicity and virulence of recent SARS-CoV-2 variant i.e. alpha, beta and delta variant, as data gathered here refers to time periods in which these were not yet identified. Fourth, we exclude pre-prints or reports published only in homepages or institutional websites.

Conclusion

Despite of heterogeneity in the included studies, two conclusions can be drawn from this review. Firstly, documenting local infection dynamics and implemented NPIs during data collection periods is crucial to understanding transmission dynamics in schools. Not all studies report these consistently. During periods of low incidence in the local population and schools with NPIs in place the risk to school staff is not necessarily higher than that of the general population and not comparable to the risk related to other high-risk professions such as health care staff. Studies reporting periods of high incidence are scarce but do show higher risk to school staff in these situations during periods where schools are not closed or NPIs are only partly in place. This may be due to the higher number and proximity of daily contacts in open schools compared to a general population under NPI public health measures.

Secondly, implementing screening and testing in schools is essential. In most of the included studies children seem less susceptible to SARS-CoV-2 infection. Students are less likely to transmit the virus to their peers or to teachers in the school setting. A large meta-analysis of prevalence studies [3] and school outbreak studies [48] supports this finding. However, these findings are biased by test strategies.

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If only symptomatic persons are tested and children show less symptoms, the number of positive cases in children is underestimated. Mass screenings of asymptomatic populations decrease the transmission of SARS-CoV-2.[49] Mass testing after index cases and frequent testing of asymptomatic students and staff was shown to reduce transmission in schools, albeit not specifically the infection risk of staff.[50] Mass testing and serial contact tracing and testing coupled with isolation and physical distancing can reduce the transmission SARS-CoV-2 in schools.[51, 52]

Implications

In Germany, schools were reopened in February 2021 despite rising population incidences (predominantly due to increased endemicity of the variant B1.1.7, now accounting for over 70% of cases in Germany). [53] A rise in cases among school-aged children is already reported by the Robert Koch Institute and the national average incidence exceeds 100 cases / 100000 / 7 days [53]. Applying the conclusions to this scenario, we expect an increasing risk to school staff and students as social contacts in open schools will outnumber out-of-school contacts in a high community NPI and infection environment. Whereas the political discourse focuses primarily on the contribution of school cases to the overall infection dynamics, the reverse dependence of the infection risks in schools on community incidences and the associated health risk to staff and students is less discussed. Presumably, the school population is misleadingly thought of as young students (only) and thus considered to be less at risk of adverse outcomes. As we have demonstrated, the staff population has to be somewhat separated from the student population in terms of infection and transmission risks. Consequently, the risk to teachers and household contacts of students and staff should be considered more prominently in the balancing political decision around school openings and closures.

With that in mind, we recommend that legislators implement well-designed mass testing and serial contact tracing and testing strategies, also including asymptomatic individuals, to minimize the risk of school outbreaks during high infection dynamics

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Legends:

- Figure 1: PRISMA flow diagram
- Table 1: PICOs for included studies
- Table 2: Characteristics of included studies
- Table 3: Studies on the risk of new infections after index case in schools (attack rate) and the risk of infection based on seroprevalence in students and teachers
- Table 4: Comparison of infection risks and attack rates of school staff with other population groups
- Supplementary table 1: Risk of bias assessment of included studies
- Supplementary table 2: Secondary transmission or attack rates in school settings
- Supplementary table 3: Infection risk in school settings
- Supplementary table 4: Infection risks and attack rates according to infection dynamics and NPIs during data collection period

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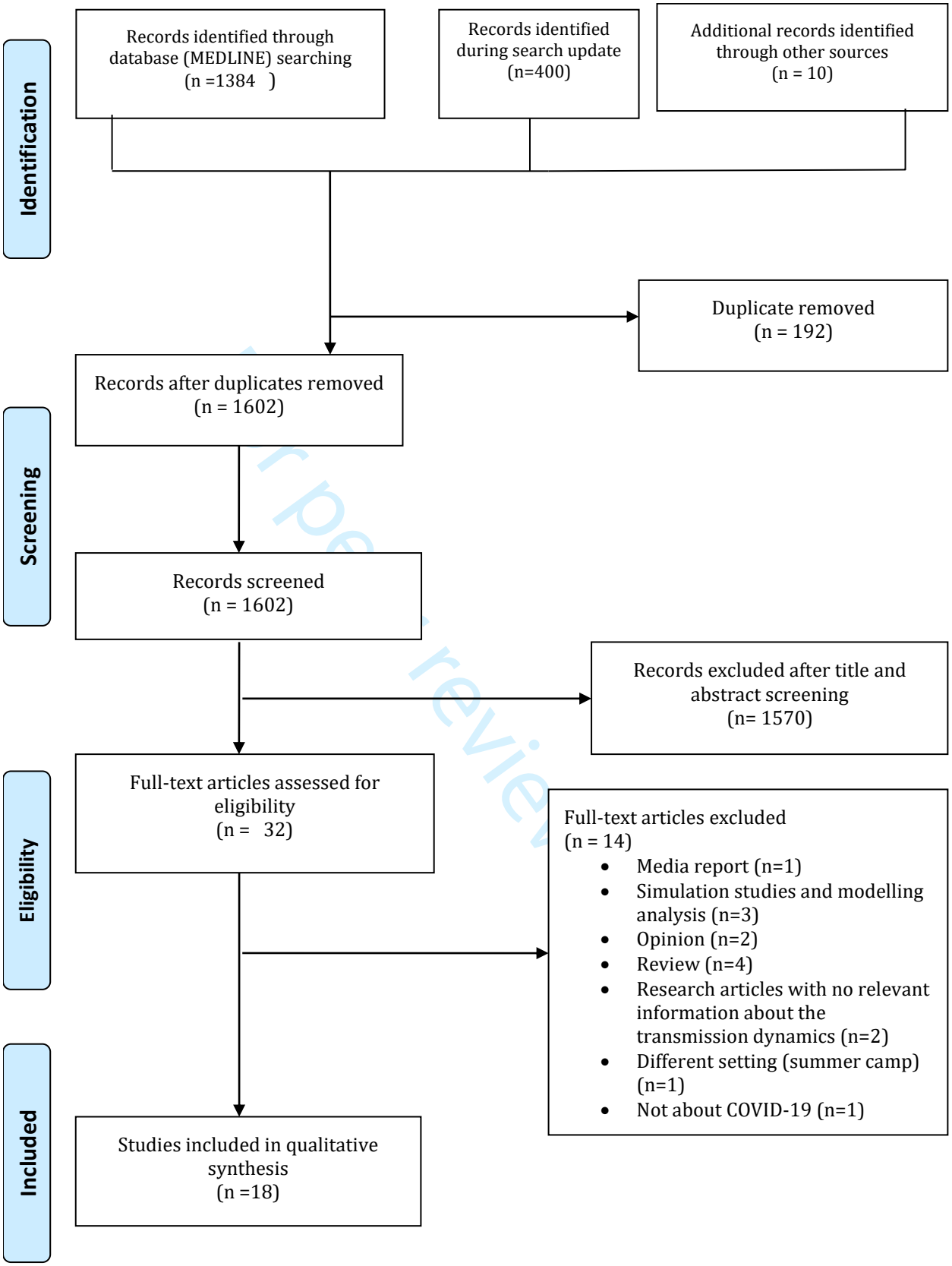


Figure 1: PRISMA (Preferred Reporting Items for Systematic Review and Meta-Analysis) flow diagram

Supplementary Table 1 Risk of Bias assessment of included studies

First author	Selection Bias	Performance Bias	Attrition Bias	Detection Bias	Reporting Bias	Information Bias	Overall
Macartney, K [30]	Medium	Low	Medium	Low	Low	Low	**Medium**
Stein-Zamir, C [32]	Low	Low	Medium	Low	Low	Low	**Medium**
Larosa, E [40]	Low	Low	Medium	Low	Low	NA	**Medium**
Public Health Agency Sweden [31]	Low	Low	Unclear	Low	Low	Unclear	**Unclear**
Vlachos, J [33]	Low	Low	Low	Low	Low	Low	**Low**
Brown, NE [26]	Medium	Unclear	High	Medium	High	High	**High**
Hernandez, A [29]	Low	Low	Medium	Low	Low	Low	**Medium**
Lopez, L [37]	Low	Low	Low	Low	Low	Low	**Low**
Heavey, L [42]	Medium	Low	Low	Low	Medium	Low	**Medium**
Yung, CF [34]	Low	Low	Low	Low	Low	Low	**Low**
Armann, JP [25]	Low	Low	Low	Low	Low	Low	**Low**
Fontanet, A [28]	Low	Low	Low	Low	Low	Low	**Low**
Danis, K [27]	Low	Medium	Low	Low	Low	Medium	**Medium**
National Centre for Immunisation Research and Surveillance [35]	Medium	Low	Medium	Low	Low	Low	**Medium**
National Centre for Immunisation Research and Surveillance [38]	Medium	Low	Medium	Low	Low	Low	**Medium**
Ismail, SA [36]	Medium	Low	Low	Low	Low	Low	**Medium**
Falk, A [39]	Medium	Low	Low	Low	Low	Low	**Medium**
Lynda, F [41]	Low	Low	Low	Low	Low	Low	**Low**

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Supplementary Table 2 Secondary transmission or attack rates in School setting

Source	First author	Main issue	Study type	Population(N)	N primary/ index cases	N primary cases staff	N primary cases student	N Contact staff	N contacts student	N contacts	N contact tested	Setting	Comparator	Date of data collection	Outcome	N tested positive	Overall attack rate	Attack rate teacher	Attack Rate Student	Comments	Infection dynamics[1]	Incident cases and deaths per day[2]	Control measures
Attack Rate in School																							
Lancet Child and Adolescent Health	Macartney , K [30]	transmission of COVID-19 among children and staff in schools and early childhood education and care (ECEC) setting	Perspective cohort study	1475(27+1448)	27 (12 children and 15 adults)	15	12	NA	NA	1448	633	15 Schools and 10 ECEC in New South Wales Australia	students	25 January - 1 May, 2020	Secondary attack rate	**18**	**1.2 %**	**child to staff = 1%** **staff to staff = 4.4%**	**child to child = 0.3%**	**18 secondary** infections: 10 infections were from children and 8 infection were from adults.	**265.65** cases per million **3.61** deaths per million	**Incidence:** Start: 0.16 case/ M Peak= 19.49 cases/ M End: 0.47 cases/ M **Deaths** Start: 0.04 Peak= 0.31 End= 0	Physical distancing, hygiene measures and educational facility cleaning Online classes from 23rd March.
Euro Surveillance	Stein-Zamir , C [32]	COVID-19 outbreak in School in Israel. The school has grade 7-grade 12.	Observational	1314 (2+1312)	2	NA	2	151	1161	1312	1312	SARS-CoV-2 **outbreak** in school of Israel after school reopening.	Students	May-June, 2020	attack rate	**178 (Student= 153 and Staff= 25)**	**13.5 %**	**16.6 %**	**13.2%**	The number of cases was higher in lower grades (grade 7-9)	1089.61 cases per million 11.32 death per million	**Incidence** Start =13.40 Peak= 90.81 End= 90.81 **Deaths** Start= 0.46 Peak= 0.69 End= 0.12	Hygiene, facemasks, social distancing and minimal interaction between classes.

[1] Total number of confirmed COVID-19 cases and deaths from COVID-19 during the data collection/ study duration in the country of study
[2] The number of incidence case per million and number of death per million per day. The data presented are at the start of the study period, peak during the study period and at the end of the study period.

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Euro Surveillance	Larosa, E [40]	Secondary transmission of COVID-19 cases in school and preschool setting of northern Italy	Observational	1248 (48+1200)	48	5	43	204	996	1200	1198 (994 student & 204 staff)	Total 36 different educational setting (school, preschool and infant toddler center) in northern Italy	Students of different school level	1 September- 15 October, 2020	attack rate	**38**	**3.2 %**	**0**	**Overall = 3.8% Secondary School AR= 6.6% Elementary school AR= 0.38% Preschool AR= 0**	All the secondary cases were among the students who were tested	1842.70 cases per million 14.75 deaths per million	**Incidence** Start= 16.13 Peak= 145.6 End= 145.60 **Deaths** Start= 0.13 Peak= 1.37 End= 1.37	Physical distancing measures, mandatory wearing of surgical masks for children, single desk and desk must me 1 m apart, extracurricular activities were suspended.
Frontiers of Public Health	Hernandez, A [29]	Secondary transmission from the school staff	Case report	202(2+200)	2	2	NA	NA	NA	200 approximately	20 (10% of the contact)	School of Panama	Students	February- March, 2020	attack rate	**7**	**3.5 %**	**3.5%**	**0**	All 15 close contacts of teachers tested negative for SARS-CoV-2 **Positive cases were the school contacts**	273.48 cases per million 6.72 deaths per million	**Incidence** Start= 0.23 Peak= 44.50 End= 44.50 **Death** Start= 0.23 Peak= 1.62 End = 1.39	School closure from March
Euro Surveillance	Heavey, L [42]	Secondary transmission in school setting	Observational	1031 (6+1025)	6	3	3	NA	NA	1025	NA	School in Ireland	NA	12 March 2020	Secondary transmission, SAR	**0 in School** **2 outside school**	**0**	**0**	**0**	**2 contacts of adult cases outside of school developed SARS-**	8.71 cases million 0.2 deaths per million	No cases and no deaths on the particular day in	School closure from 12th March

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																			CoV-2 infection with overall AR 0.17%		Ireland	
Clinical Infectious Diseases	Yung, CF [34]	Secondary transmission in school setting	Observational	122 (3+119)	3	1	2	NA	NA	119	119 (8+34+77)	3 different school in Singapore	NA	February-March 2020	Secondary transmission, AR	**0**	**0**	**0**	**0**	No secondary cases were found among close contacts.	**155.55 cases per million** **0.17 deaths per million**	**Incidence** Start= 0.51 Peak= 12.48 End= 8.03 **Deaths** Start= 0.34 Peak= 0.34 End= 0	Terminal cleaning of school, suspension of extracurricular activities, preschool was closed for 14 days
Clinical Infectious Diseases	Danis, K [27]	Tertiary transmission in school setting	Observational	113 (1+112)	1	NA	1	NA	NA	112	101	3 schools and 1 ski club in French Alps	Student and teachers	February 2020	Tertiary transmission	**0**	**0**	**0**	**0**	No transmission in schools	1.44 cases per million 0.03 deaths per million	**Incidence** Start= 0.02 Peak= 0.31 End= 0.29 **Deaths** Start= 0.02 Peak= 0.02 End= 0	**Use of surgical masks**
National Centre for Immunisation Research and Surveillance, NSW, Australia	NCIRS [35]	secondary transmission in differed educational setting	Observational	527 (6+521)	6	2	4	62	459	521	319	6 different educational (5 schools and 1 ECEC)	Student; teacher	10 April – 3 July, 2020	Secondary transmission	**0**	**0**	**0**	**0**	No secondary transmissions	80.20 cases per million 1.96 deaths per million	**Incidence** Start= 4.20 Peak= 7.61 End=7.61 **Deaths** Start= 0.12 Peak= 0.31 End= 0	**Schools** were re-opened, full face-to-face teaching commenced while implementing COVID-19 safe practice

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National Centre for Immunisation Research and Surveillance, NSW, Australia	NCIRS [38]	secondary transmission in differed educational setting	Observational	3863(33+3824)	39	7	32	385	3439	3824	3641	34 different educational setting in NSW which includes 28 schools and 6 ECEC	student	4 July – 25 September, 2020	Secondary transmission; attack rate	**33**	**0.86 %**	**1.29 %**	**0.81%**	Among tested: AR= 0.9% High School AR= 1.1% Primary school AR= 0.4% ECEC AR= 0.7%	728.36 cases per million 30.04 deaths per million 744 cases across NSW. Among them 555 were locally acquired.	**Incidence** Start= 7.18 Peak= 28.08 End= 0.63 **Death** Start= 0 Peak= 2.31 End= 0.04	Schools were open for face-to-face learning while implementing COVID-19 safe practice.
Lancet Infectious Diseases	Ismail, SA [36]	estimating the rate of SARS-CoV-2 infection and outbreaks among staff and students in educational settings during the summer half-term	Prospective cross sectional analysis	928000 (Students only)	55	34	21	NA	NA	NA	NA	Schools in England	Student	June/July 2020	Secondary transmission	**NA**	**122**	**100**	**22**	Considering only outbreak situation there were 122 secondary cases in school during this period. Among them 100 cases were staff and 22 were students	661.80 cases per million 53.63 deaths per million	**Incidence** Start= 21.27 Peak= 21.65 End= 13.01 **Death** Start= 1.27 Peak= 3.80 End= 0.29	School closed from March 20 and reopened from June 1st after implementing smaller classes separated into distant social bubbles
Morbidity and Mortality	Falk, A [39]	Secondary transmission in	Observational	5530 (184+5346)	184	58	126	596	4750	5346	5530	Schools in Wiscon	Student	August-29 November,	Secondary transmission;	**191**	**0.13 %**	**0**	**0.14%**	Among 191 tested positive,	26,802.94 cases	**Incidence**	Compulsory masks for

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Weekly Report		school setting										sin, USA		2020	attack rate					7 were secondary cases There was no secondary transmission among staff	per million 339.08 deaths per million	Start= 169.71 Peak= 628.19 End= 423.22 **Deaths** Start= 3.31 Peak= 6.86 End= 3.13	student and staff. Classroom size 11-20 students and indoor lunch.

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Supplementary Table 3 Infection risk in School setting

Source	First author	Main issue	Study type	Population (N)	N primary/ index cases	N primary cases staff	N primary cases student	N Contact staff	N contacts student	N contacts	N tested	Setting	Comparator	Date	Outcome	N tested positive	Overall risk of infection	Infection risk in teacher	Infection risk in student	Comments	Infection dynamics[1]	Incident cases and deaths per day[2]	Control measures
Infection risk in School																							
Public Health Agency Sweden	Public Health Agency Sweden [31]	Risk of SARS-CoV-2 infection in teachers in Sweden	Report (observational study)	Teachers working in different school in Sweden	364760	479 cases	NA	NA	NA	NA	NA	School of Sweden	Teachers of different school forms	March-May 2020	Relative risk	**NA**		**RR and 95%CI** High School teachers **0.7(0.5-1)** Primary school teachers **1.1(0.9-1.3)** Leisure educators **0.8(0.5-1.3)** Preschool teachers **0.7(0.6-0.9)** Other educators with theoretical specialists	NA	The relative risk is higher for primary school teachers and student assistants compared to other groups working in schools in Sweden	3,800.47 cases per million 435.08 deaths per million	**Incidence** Start= 0 Peak= 80.01 End= 26.24 **Death** Start= 0.1 Peak= 18.32 End= 0	Partial school closure

[1] Total number of COVID-19 cases and deaths from COVID-19 per million during the period of data collection/study duration in the country of study
[2] The number of incidence case per million and number of death per million per day. The data presented are at the start of the study period, peak during the study period and at the end of the study period

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																		1(0.7-1.5) Nanny **1(0.8-1.2)** Student assistants and other **1.1(0.8-1.4)**					
medRvix	Vlachos,J [33]	Risk of COVID-19 infection in the parents, high school teachers, lower secondary school teacher and their partners	Retrospective data analysis	Upper secondary and lower secondary teachers, their partners and student's parents from Sweden	NA	NA	NA	NA	NA	NA	NA	School and household of Sweden	Teachers of different school form, their partners And parents of students	June 15, 2020	Odds ratio and 95%CI	**NA**	**lower secondary school** **parents** **1.15 (1.03-1.27)** **Partners** **1.30(1.00-1.68)**	**2.01(1.52 – 2.67)**	NA	Secondary school is the reference group	2,833.37 cases per million 92.09 deaths per million	**Incidence** Start= 64.16 Peak= 168.13 End= 79.19 **Deaths** Start= 0.79 Peak= 10.69 End= 5.25	Partial school closure
Emerging Infectious Diseases	Brown, NE[26]	Teacher to student transmission	Observational (Serological survey)	121 (1+120)	1	1	NA	NA	120	120	21	School in USA.	Students	Blood sample collection date: 13 March, 2020	Infection rate	**2**	**1.66%**	**0**	**1.66%**	blood collection was done 14 days after exposure	587.87 cases per million 16.13 deaths per	**Incidence** Start= 0.02 Peak= 79.50 End= 79.50 **Death**	NA

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																				million	Start= 0 Peak= 3.31 End= 3.31	
medRxiv	Lopez, L [37]	Sero-prevalence of anti-SARS-CoV-2 IgG antibodies in school staff in Midwestern USA	Population based sero prevalence study	1261 eligible School staff	NA	NA	NA	NA	NA	NA	753 participated in the survey (Annual check-up)	18 year or older staff of Lake Central School Corporation located in suburban Indiana	General population	July 2020	Seroprevalence of antibody	22 participants tested positive for SARS-CoV-2 antibodies	2.9% (95%CI 1.8-4.4%) **When Corrected for performance parameter** 1.7%	1.7%		The seroprevalence of LCSC staff is 1.7% which is below the seroprevalence reported in metropolitan areas throughout the USA	5,658.69 cases per million 78.18 deaths per million	**Incidence** Start= 156.52 Peak= 229.03 End= 207.54 **Death** Start= 2.14 Peak = 4.33 End= 3.72	School closure, While entering the building for data collection facemask, social distancing and hand sanitizer were mandatory
medRxiv	Armann, JP [25]	Seroprevalence study in student and teacher to assess their role SARS-CoV-2 transmission	Seroprevalence study	2045 (1538 students and 507 teachers)	NA	NA	NA	NA	NA	NA	2045	Secondary school in Saxony, Germany	students	25 May, 2020 and 30 June, 2020	Prevalence (Seroprevalence)	12 (11 students and 1 teacher)	0.6%	0.2%	0.7%	Seroprevalence study in 13 different secondary school in Saxony, Germany	374.07 cases per million 26.90 deaths per million	**Incidence** Start= 12.75 Peak= 15.13 End= 4.49 **Death** Start= 1.35	School closure, face mask and social distancing

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																					Peak= 3.37 End= 0.17	
medRvix	Fontanet, A [28]	to assess the infection attack rate using serological assays	Retrospective cohort study (Seroprevalence study)	661 (including 240 pupil, 211 parents, 127 sibling, 53 teacher; 27 non-teaching staff and 3 other)	NA	NA	NA	NA	NA	NA	661	Secondary school and household	Student, parents	30 March -4 April, 2020	Infection attack rate	171 (pupil= 92; teacher= 23; school staff= 16; parents=24; sibling = 13, and other= 3)	25.9% (22.6-29.4)	12.18%	28.75%	Seroprevalence study in secondary school and household in Oise in Northern France	41.81 cases per million 69.49 deaths per million	**Incidence** Start= 66.70 Peak= 116.88 End=13.10 **Death** Start= 6.40 Peak= 19.18 End= 16.13	NA
medRvix	Lynda, F[41]	Risk of COVID-19 and risk of hospitalization with COVID-19 among teachers compared with healthcare workers and general population of working age	Case control study	871568	NA	NA	NA	NA	NA	NA	NA	School, hospital and household of Scotland	Healthcare worker, Household member of healthcare worker and teacher	March 2020-January 2021	Risk rate/ odds ratio	NA		**Before re-opening**: Any cases 0.48 (0.39-0.60); Hospitalization 0.51 (0.31-0.82) **School re-opened:** Any cases 1.42	**Before reopening,** **Any case** HH_Teacher 0.78 (0.66-0.91) HCW 8.75 (8.29-9.23); HH_HCW 2.32 (2.11-2.56) **Hospitalization** HH_Teacher 0.63	comparators are healthcare workers and household member of teachers and healthcare worker The results presented are adjusted ratios obtained	56389.96 cases per million 1566.83 deaths per million	**Incidence** Start= 0.03 Peak= 1004.51 End= 312.14 **Death** Start= 0.01 Peak= 26.90 End= 17.75	School close and again reopen

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																		(1.35-1.49); Hospitalization 0.97 (0.72-1.29)	(0.44-0.91); HCW 3.15 (2.71-3.66); HH_HCW 1.87 (1.52-2.30) **Re-opened** **Any cases** HH_Teacher 1.01 (0.95-1.06); HCW 2.41 (2.33-2.49); HH_HCW 1.25 (1.20-1.31) **Hospitalization** HH_Teacher 0.97 (0.74-1.27); HCW 1.82 (1.55-2.14), HH_HCW 1.04 (0.83-1.23)	from logistic regression			

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Supplementary Table 4 Infection risks and attack rates according to infection dynamics and Non-pharmaceutical Interventions (NPIs) during data collection period

Infection dynamics cases	Infection dynamics deaths	control measures	Study	Infection risks for students	Attack rates for students	Infection risks for teachers	Attack rate for teachers	Comparator of infection risk for students/teachers	Comparator for teachers/teachers	Comment
Total number of cases in country / million / week < 100	N of deaths per million in data collection period < 100	School open	Heavey, L [42]		0		0			No secondary transmission in school setting
		School open with NPIs	Danis, K [27]		0		0			No secondary transmission in school setting
			National Centre for Immunisation Research and Surveillance, NSW, Australia (2nd term) [35]		0		0			No secondary transmission in school setting
		NA	Fontane, A [28]	12.18 %		28.75%				
Total number of cases in country / million / week > 101 and < 1000	N of deaths per million in data collection period < 100	School open	Hernandez, A [29]		0		3.5 %			
			Yung, CF [34]		0		0			No secondary transmission
		School open with NPIs	Macartney, K [30]		0.3 %			Child to staff transmission rate: 1%	Staff to staff transmission rate 4.4%	

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		School close/ online teaching	Armann, JP [25]	0.7 %		0.2%				
		School open with NPIs	National Centre for Immunisation Research and Surveillance, NSW, Australia (3rd term) [38]		0.81%		1.29%			
			Larosa, E [40]		3.8 %		0	Secondary school= 6.6%		
								Elementary school = 0.38%		
		School open with NPIs	Ismail, SA [36]		N= 22		N= 100			Absolute number of secondary infections during outbreak in school
		NA	Brown, NE [26]	1.66%		0				
Total number of cases in country / million / week > 101 and < 1000	N of deaths per million in data collection period > 50 and <100	School open with NPIs	Falk, A [39]		0.14%		0			
			Stain-Zamir, C [32]		13.2%		16.6%			
		Partial school closure	Public health Agency, Sweden [31]						High school= 0.7(0.5-1)	Relative risk among teachers
									Primary school=	

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									1.1(0.9-1.3)	
									Preschool= 0.7 (0.6-0.9)	
									Student assistant= 1(0.8-1.4)	
			Valchos, J [33]						OR=2.01 (1.52-2.67)	Secondary school teachers are a reference group
		School closes/ online	Lynda, F [41]	NA	NA	0.48(0.39-0.60)				
			Lopez, L [37]	0		1.7%				Seroprevalence assay done only in staff
		School reopen with NPI	Lynda, F [41]	NA	NA	1.42 (1.35 - 1.49)				

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Reporting checklist for systematic review and meta-analysis.

Based on the PRISMA guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMAreporting guidelines, and cite them as:

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	Reporting Item	Page Number
Title		
#1	Identify the report as a systematic review, meta-analysis, or both.	1
Abstract		

Structured summary	#2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number	2
Introduction			
Rationale	#3	Describe the rationale for the review in the context of what is already known.	4
Objectives	#4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	NA
Methods			
Protocol and registration	#5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address) and, if available, provide registration information including the registration number.	5
Eligibility criteria	#6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rational	5
Information sources	#7	Describe all information sources in the search (e.g., databases with dates of coverage, contact with study authors to identify additional studies) and date last searched.	5
Search	#8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5

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Study selection	#9	State the process for selecting studies (i.e., for screening, for determining eligibility, for inclusion in the systematic review, and, if applicable, for inclusion in the meta-analysis).	5-6
Data collection process	#10	Describe the method of data extraction from reports (e.g., piloted forms, independently by two reviewers) and any processes for obtaining and confirming data from investigators.	6
Data items	#11	List and define all variables for which data were sought (e.g., PICOS, funding sources), and any assumptions and simplifications made.	6
Risk of bias in individual studies	#12	Describe methods used for assessing risk of bias in individual studies (including specification of whether this was done at the study or outcome level, or both), and how this information is to be used in any data synthesis.	6
Summary measures	#13	State the principal summary measures (e.g., risk ratio, difference in means).	6
Planned methods of analyis	#14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I2) for each meta-analysis.	NA
Risk of bias across studies	#15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	NA
Additional analyses	#16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which	NA

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		were pre-specified.	
Results			
Study selection	#17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7
Study characteristics	#18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citation.	9-10
Risk of bias within studies	#19	Present data on risk of bias of each study and, if available, any outcome-level assessment (see Item 12).	7-8
Results of individual studies	#20	For all outcomes considered (benefits and harms), present, for each study: (a) simple summary data for each intervention group and (b) effect estimates and confidence intervals, ideally with a forest plot.	NA
Synthesis of results	#21	Present the main results of the review. If meta-analyses are done, include for each, confidence intervals and measures of consistency.	11-13
Risk of bias across studies	#22	Present results of any assessment of risk of bias across studies (see Item 15).	7-8
Additional analysis	#23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	NA
Discussion			

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Summary of Evidence	#24	Summarize the main findings, including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., health care providers, users, and policy makers	14-15
Limitations	#25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review level (e.g., incomplete retrieval of identified research, reporting bias).	14-15
Conclusions	#26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	15
Funding			
Funding	#27	Describe sources of funding or other support (e.g., supply of data) for the systematic review; role of funders for the systematic review.	17

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